UNIVERSITÄT AUGSBURG

Master's Thesis

Circle Bundles with Positive Scalar Curvature over Enlargeable Manifolds

Johannes Nikolaus Vinnemeier

Supervisors: Prof. Dr. Bernhard Hanke,
Dr. Georg Frenck

Faculty of Mathematics, Natural Sciences and Materials Engineering

University of Augsburg

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Introduction

The scalar curvature of a Riemannian Manifold (M, g) is a smooth function $\mathrm{scal}\colon M \to \mathbb{R}$ measuring how much the volume of small geodesic balls around a point deviates from that of euclidian balls. While it is the weakest curvature condition for a Riemannian manifold, the presence of positive scalar curvature can still have strong topological implications. The archetypal example is the celebrated Gauß-Bonnet Theorem:

Theorem (Gauß-Bonnet)**.** *Let (M, g) be a closed Riemannian 2-manifold. Then*

$$\int_M \mathrm{scal}\,\mathrm{d}A = 4\pi\chi(M),$$

where $\chi(M)$ is the Euler characteristic of M.

So any closed surface carrying a positive scalar curvature metric must be diffeomorphic to either the sphere S^2 or $\mathbb{R}P^2$. For non-compact complete manifolds, merely having positive scalar curvature may be too weak of a condition. The problem is that scal can still tend to zero quite rapidly at infinity. However, if the rate of decay at infinity can be controlled, positive scalar curvature still has profound consequences. For example, Chen has recently shown a decomposition result for 3-manifolds under the assumption that a positive scalar curvature metric with controlled decay at infinity exists [Chen2025]. This naturally raises the question of how fast the scalar curvature of a given complete Riemannian manifold X decays at infinity. One statement in this direction is the following theorem by Gromov:

Theorem (Enlargeable Decay Theorem, [Gro18], page 15f.)**.** *Let X be a complete Riemannian manifold, $V \to M$ a two dimensional vector bundle over an enlargeable manifold M. Assume that*

(i) There exists a proper map $f\colon X \to V$, $\deg(f) \neq 0$,

(ii) The total space of the circle bundle corresponding to V is also enlargeable.

Then

$$\min_{B_{x_0}(R)} \mathrm{scal}(X) \leq \frac{8\pi^2}{(R - R_0)^2}$$

for some $R_0 = R_0(X, x_0)$ and all $R > R_0$.

It is easy to see that condition (ii) is always fullfilled if the bundle $V \to M$ is trivial (compare Lemma 3.3 (i)), which lead Gromov to pose the following question:

Question ([Gro18])**.** "What happens to nontrivial bundles $V \to M$? [...] In general, the examples in [BH10] indicate a possibility of non-enlargeable circle bundles over enlargeable M; yet, it seems hard(er) to find such examples, where the corresponding total space would admit metrics with $\mathrm{scal} > 0$."

As it turns out, such examples do in fact exist, that is, circle bundles over enlargeable manifolds, where the total space admits a metric of positive scalar curvature and thus, in particular, is not enlargeable. Bundles with this behaviour have recently been constructed in a paper by Kumar and Sen [KS25] by utilizing tools from symplectic geometry. A very rough sketch of the construction can be given as follows. Take M to be a so-called Donaldson divisor of a suitable closed symplectic manifold, e.g., the torus T^{2n}, $n \geq 3$. Such a Donaldson divisor is a certain codimension 2 symplectic submanifold $M^{2n-2} \subseteq T^{2n}$, which in some cases can be shown to be enlargeable. Now define a circle bundle $E := \partial\nu(M)$ over M as the boundary of a tubular neighbourhood $\nu(M) \subseteq T^{2n}$ of M. In order to construct a positive scalar curvature metric on E, consider the nullbordism $W := T^{2n} - \nu(M)$ obtained by removing the tubular neighbourhood from the ambient manifold. Now explicitly construct a Morse function $\psi\colon W \to \mathbb{R}$ with critical points of index at most n, which shows that E is obtained from the empty set by surgeries of codimension at least $n \geq 3$. Thus, E admits a metric of positive scalar curvature by the Gromov-Lawson-Schoen-Yau Surgery Theorem (see Theorem 2.7).

In this thesis, we provide an alternative, much more direct construction relying simply on basic topological concepts. It should be said, however, that this alternative approach only provides examples in the totally non-spin case.

The thesis is organized as follows. Chapter 1 recalls some concepts from the broader theory of principal bundles, laying the foundation for the subsequent chapters. In Chapter 2, we briefly introduce the reader to the theory of positive scalar curvature and prove two existence theorems for positive scalar curvature metrics, which are used in the main construction in Chapter 4. The proofs of these theorems are based on methods from surgery theory, which we also take some time to explain. Chapter 3 provides a brief discussion of the concept of enlargeability, which is the central idea underlying this thesis. Finally, Chapter 4 contains the main results. We present a simple construction of circle bundles over enlargeable manifolds whose total spaces admit metrics of positive scalar curvature in all dimensions ≥ 4. These bundles are then used to obtain counterexamples to the Enlargeable Decay Theorem above if condition (ii) is dropped.

1 Preliminaries

This chapter develops the necessary preliminaries for what follows, by recalling the notions of principal bundles, classifiying spaces and characteristic classes. We assume the reader to be familiar with the concept of a fiber bundle with structure group. A classical reference for this is for example [Ste51]. Moreover, we presuppose some basic familiarity with homotopy and homology theory.

1.1 Principal Bundles and Associated Bundles

Definition 1.1. Let G be a topological group. A G-principal bundle is a fiber bundle with fiber G and structure group G acting by left translations.

The total space of a G-principal bundle $P \xrightarrow{\pi} B$ carries a natural right action of G defined as follows. Around $y \in P$ choose a local trivialisation

$$\phi\colon U \times G \to \pi^{-1}(U)$$

and let $\phi^{-1}(y) = (\pi(y), h)$. Then set

$$y \cdot g := \phi(\pi(y), hg).$$

This is well defned since by definition G acts on itself by *left* translations under a change of local trivialisation. A morphism of principal bundles should respect this right action, thus we define:

Definition 1.2. A morphism from a G-principal bundle $P \to B$ to an H-principal bundle $P' \to B'$ is a pair (F, f) of continuous maps $F\colon P \to P'$ and $f\colon B \to B'$ together with a homomorphism $\rho\colon G \to H$, such that

$$\begin{tikzcd} P \arrow[r, "F"] \arrow[d] & P' \arrow[d] \\ B \arrow[r, "f"] & B' \end{tikzcd}$$

commutes and such that F is ρ-equivariant, that is $F(x \cdot g) = F(x) \cdot \rho(g)$ for all $x \in P$ and $g \in G$. If $G = H$ we assume ρ to be the identity and if $B = B'$ we assume f to be the identity.

The condition of equivariance in the above definition is actually quite strong as illustrated by the following proposition.

Proposition 1.3. *Any morphism of G-principal bundles over the same base space is an isomorphism.*

Proof. First suppose we are given a morphism

$$F\colon B \times G \to B \times G$$

of trivial bundles. Then by definition F is of the form

$$F(x, g) = (x, \phi_x(g))$$

for a continuous family of equivariant maps $\phi_x\colon G \to G$. By equivariance $\phi_x(g) = \phi_x(e)g$ and thus an inverse of F is given by

$$F^{-1}(x, g) := (x, \phi_x(e)^{-1}g).$$

Since every G-principal bundle is locally trivial this shows that we can always find local inverses in the general case and since inverses are unique they fit together to define a global inverse. $\square$

To any fiber bundle with structure group G one can canonically associate a G-principal bundle, obtained by applying a general construction, which can be used to change the fiber of a given bundle. We now recall this construction of changing the fiber. Let $p\colon E \to B$ be a fiber bundle with fiber F and structure group G and let F' be another space upon which G acts effectively. Then we can construct a new bundle as follows. Choose an open cover $(U_\alpha)_{\alpha \in A}$ of B together with local trivialisations $\phi_\alpha\colon U_\alpha \times F \to p^{-1}(U_\alpha)$. Let $\tau_{\alpha\beta}\colon U_\alpha \cap U_\beta \to G$ be the corresponding transition functions defined by

$$(\phi_\beta^{-1} \circ \phi_\alpha)(x, f) = (x, \tau_{\alpha\beta}(x) \cdot f).$$

Define

$$E' := \left(\coprod_{\alpha \in A} U_\alpha \times F' \right) \Big/ \sim$$

where the equivalence relation $\sim$ is defined by

$$(b, f', \beta) \sim (b, \tau_{\alpha,\beta}(b) \cdot f', \alpha)$$

and let $p' \colon E' \to B$, $[(b, f', \alpha)] \mapsto b$. Then the maps

$$\psi_\alpha \colon U_\alpha \times F' \to (p')^{-1}(U_\alpha),\ (b, f') \mapsto [(b, f', \alpha)].$$

define local trivialisations of p' with the same transition functions $\tau_{\alpha\beta}$, giving $p' \colon E' \to B$ the structure of a fiber bundle with fiber F' and structure group G. We say that this new bundle is obtained from $E \to B$ by *changing the fiber* from F to F' (with the G-action on F' being implicit). Note that we can also obtain $E \to B$ from $E' \to B$ by changing the fiber from F' back to F. Thus no information is lost in this process. Considering the special case $F' = G$ with G acting on itself by left translations, we call the bundle obtained from $E \to B$ by changing the fiber from F to G the *associated principal bundle*. Morally speaking if we want to understand the bundle $E \to B$ it is sufficient to understand its associated prinicpal bundle, as it contains the same information as we have seen above. The benefit of studying principal bundles is that they are highly symmetric objects, and thus usually easier to understand. For example many constructions simplify for principal bundles, such as the concept of universal bundles as we will see in the next subsection. Another such instance is the construction for changing the fiber from above, which can be carried out more directly in the case of principal bundles and has the benefit of also allowing for non-effective group actions. This is known as the *Borel Construction*.

Lemma 1.4. *Let $p \colon P \to B$ be a G-principal bundle and F a space upon which G acts from the left. Then*

$$E := (P \times F)\big/\sim, \quad (y, f) \sim (y \cdot g, g^{-1} f)$$

together with the map $\pi \colon E \to B$ given by $[(y, f)] \mapsto p(y)$ is a fiber bundle with fiber F and structure group G. Moreover, if the action of G on F is effective, this bundle is isomorphic to the bundle obtained from $P \to B$ by changing the fiber from G to F using the construction outlined above.

Proof. To see that $P \times_G F$ is locally trivial let $\psi \colon U \times G \to P$ be a local trivialisation. Then the homeomorphism

$$\psi \times \mathrm{id}_F \colon U \times G \times F \to p^{-1}(U) \times F$$

descends to a homeomorphism

$$\tilde{\kappa} \colon (U \times G \times F)\big/\sim \to (p^{-1}(U) \times F)\big/\sim = \pi^{-1}(U),$$

where the equivalence relation on $U \times G \times F$ is given by $(x, g, f) \sim (x, gh, h^{-1} f)$. Moreover the map

$$U \times G \times F \to U \times F,\ (x, g, f) \mapsto (x, gf)$$

descends to a homeomorphism $(U \times G \times F)\big/\sim \to U \times F$ with inverse given by $(x, f) \mapsto [(x, e, f)]$. Under this homeomorphism $\tilde{\kappa}$ corresponds to a homeomorphism $\kappa \colon U \times F \to \pi^{-1}(U)$, which is explicitly given by $(x, f) \mapsto [(\psi(x, e), f)]$ and

$$\begin{tikzcd} U \times F \arrow[r, "\kappa"] \arrow[dr, "\text{proj.}"'] & \pi^{-1}(U) \arrow[d, "\pi"] \\ & U \end{tikzcd}$$

commutes. So $P \times_G F$ is locally trivial. To see that this construction agrees with the change of fiber construction from above if G acts effectively we have to compute the transition functions of the local trivialisations. Let $U_\alpha, U_\beta \subseteq B$ be two trivialising neighbourhoods for P with local trivialisations ψ_α, ψ_β and transition function $\tau_{\alpha,\beta} \colon U_\alpha \cap U_\beta \to G$. Moreover let $\kappa_\alpha, \kappa_\beta$ be the corresponding local trivialisations for $P \times_G F$ as constructed above. Then we compute

$$(\kappa_\alpha^{-1} \circ \kappa_\beta)(x, f) = \kappa_\alpha^{-1}([(\psi_\beta(x, e), f)]) = (x, \tau_{\alpha,\beta}(x) f).$$

Thus the statement follows from the fact that fiber bundles with structure group are uniquely determined by the transition functions of local trivialisations. $\square$

In view of the above constrcution one usually denotes the bundle obtained from $P \to B$ by changing the fiber from G to F as $P \times_G F \to B$. The Borel construction is natural in the following sense.

Lemma 1.5. *Let $p\colon P \to B$ be a G-prinicpal bundle, F a space upon which G acts from the left and $f\colon B' \to B$ a continuous map. Then*

$$(f^*P) \times_G F \cong f^*(P \times_G F).$$

Proof. An explicit model for the pullback bundle is

$$f^*P := \{(y, b') \in P \times B' \mid p(y) = f(b')\}.$$

Then

$$(f^*P) \times_G F = \{(y, b', f) \in P \times B' \times F \mid p(y) = f(b')\}/\sim,$$

for $(y, b', f) \sim (y \cdot g, b', g^{-1} \cdot f)$. Moreover

$$\begin{aligned} f^*(P \times_G F) &= \{([y, f], b') \in P \times_G F \times B' \mid p(y) = f(b')\} \\ &\cong \{(y, f, b') \in P \times F \times B' \mid p(y) = f(b')\}/\sim \end{aligned}$$

for $(y, f, b') \sim (y \cdot g, g^{-1} \cdot f, b')$. Clearly these are isomorphic bundles. □

Remark 1.6. We explicitly remark that the action of G on F in the above Lemma need not be effective. If we restrict ourselves to effective actions, a more general naturality result holds for changing the fiber of arbitrary fiber bundles. Namely first pulling a bundle back under some continuous map and then changing the fiber amounts to the same as first changing the fiber and then pulling back. This can easily be seen by comparing the transition functions of local trivialisations in both cases and observing that they agree.

Lets look at some concrete examples for the theory developed above.

Example 1.7. For simplicity we assume all base spaces B to be smooth manifolds in this example.

(i) Let $E \to B$ be a real rank n vector bundle. Choosing a bundle metric gives $E \to B$ the structure of a fiber bundle with fiber $\mathbb{R}^n$ and structure group $O(n)$. The associated principal bundle is explicitly given by the *orthonormal frame bundle* $\mathrm{O}(E)$ *of* E, whose fiber over a point $x \in B$ is the space of all ordered orthonormal bases of E_x. The topology on $O(E)$ is the subspace topology $O(E) \subseteq E \times \ldots \times E$. Local trivialisations of the canonical map $p\colon \mathrm{O}(E) \to B$ can be defined as follows. Choose an orthonormal local frame $e_1, \ldots, e_n$ of E defined over an open set $U \subseteq B$. Then a trivialisation $\phi\colon U \times \mathrm{O}(n) \to p^{-1}(U)$ is given by

$$(x, A) \mapsto \left(\sum_{i=1}^n a_{i1} e_i(x), \ldots, \sum_{i=1}^n a_{in} e_i(x) \right),$$

where $A = (a_{ij})$, i.e. $\phi(x, A)$ is the basis of $E_x = p^{-1}(x)$, whose coordinates in the frame (e_i) are precisely the columns of the matrix A. The transition functions of these trivialisations agree with the transition functions of the trivialisations of E. Again, since the transition functions uniquely determine the bundle, we conclude that $O(E)$ is in fact isomorphic to the associated principle bundle as defined at the beginning of this section. Analogously, if $E \to B$ is a complex rank n vector bundle, choosing a hermitian metric on E allows us to reduce the structure group to $U(n)$ and the corresponding associated principal bundle is the *unitary frame bundle.*

(ii) Let $p\colon \tau_1^n \to \mathbb{C}P^n$ be the tautological complex line bundle equipped with the canonical hermitian metric. Since this bundle is of rank 1, its unitary frame bundle is simply the subspace $U(\tau_1^n) \subseteq \tau_1^n$ of all vectors of norm 1, together with the restriction of p to this subspace. Recall that

$$\tau_1^n = \{(L, z) \in \mathbb{C}P^n \times \mathbb{C}^{n+1} \mid z \in L\}.$$

So explicitly

$$U(\tau_1^n) := \{(L, z) \in \tau_1^n \mid |z| = 1\}.$$

Consider the diffeomorphism

$$U(\tau_1^n) \to S^{2n+1},\ (L,z) \mapsto z$$

with inverse $S^{2n+1} \to U(\tau_1^n),\ z \mapsto (\pi(z), z)$, where $\pi\colon S^{2n+1} \to \mathbb{C}P^n$ is the canonical projection. Then the restriction of p to $U(\tau_1^n)$ corresponds to π under this diffeomorphism. Hence π defines an $U(1) \cong S^1$-principal bundle, which agrees with the associated principal bundle of $\tau_1^n \to \mathbb{C}P^n$.

(iii) An example, which will be of relevance later, is the following. Let $E \to B$ be an S^1-principal bundle over a manifold B without boundary. We can change the fiber of E from S^1 to D^2 to obtain another bundle $DE \to B$, where S^1 acts on D^2 by rotations. By definition of the change-of-fiber construction, trivialisations $U \times D^2 \to DE$, restricted to the boundary $U \times S^1 \to \partial DE$, have the same transition functions as the corresponding trivialisations of $E \to B$. Since the action of S^1 on D^2 just extends the action of S^1 on itself, we see that $E \cong \partial DE$. So, in particular, the total space of any S^1-principal bundle is null-bordant.

1.2 Universal Bundles and Classifying Spaces

A natural question is which fiber bundles with specified fiber and structure group can exist over a given base space. Again, this classification problem turns out to be easier for principal bundles. For simplicity we assume all base spaces to be CW-complexes in this section.

Definition 1.8. A G-principal bundle $E \to B$ is called *universal* if E is weakly contractible as a topological space.

The significance of this definition is established by the following theorem. We say that two morphisms of G-principal bundles are G-homotopic if they are homotopic through morphisms of G-principal bundles.

Theorem 1.9. *Let $E \to B$ be a universal G-principal bundle. Then for any other G-principal bundle $P \to X$ there exists a morphism $P \to E$ of G-principal bundles, which is unique up to G-homotopy.*

Before showing the theorem let us discuss its implications. Firstly we can equivalently formulate the theorem by saying that $E \to B$ is a terminal object in the homotopy category of G-principal bundles over CW-complexes and thus in particular uniquely determined (up to G-homotopy equivalence). This justifies writing $EG := E$ and $BG := B$ and we refer to $EG \to BG$ as *the* universal G-principal bundle. The base space BG is called the *classifiying space* of G. Since any bundle morphism $(F, f)\colon P \to EG$ induces a morphism $P \to f^*EG$ (and vice versa, which is true for arbitrary fiber bundles) we conclude from Proposition 1.3 that

$$P \cong f^*EG.$$

We call f a *classifiying map* for P. By homotopy invariance of the pullback we get a well-defined bijection

$$\phi_X\colon [X, BG] \to \mathcal{P}_G(X),\ [f] \mapsto [f^*EG],$$

where $\mathcal{P}_G(X)$ denotes the set of isomorphism classes of G-principal bundles over X. To see that ϕ_X is indeed bijective note that the existence result in Theorem 1.9 shows that ϕ_X is surjective and the uniqueness result shows that it is injective. Moreover ϕ_X is clearly natural in X. Thus we have shown.

Corollary 1.10. *The maps ϕ_X define a natural isomorphism of contravaraint functors*

$$[-, BG], \mathcal{P}_G(-)\colon \mathsf{HCW} \to \mathsf{Set},$$

where HCW *denotes the homotopy category of CW-complexes.*

Theorem 1.9 will be a simple application of the following theorem.

Theorem 1.11. *Let $P \to X$ be a G-principal bundle and $A \subseteq X$ a subcomplex. Then any morphism $P|_A \to EG$ of G-principal bundles can be extended to a morphism $P \to EG$ of G-principal bundles.*

The proof of this Theorem based on an inductive cell by cell approach. We refer the reader to [Ebe12, Theorem 2.7.1].

Proof of Theorem 1.9. Applying Theorem 1.11 to $A = \emptyset$ shows the existence of a morphism $P \to EG$. Now suppose we are given morphisms $F_0, F_1 \colon P \to EG$. Consider the G-principal bundle

$$\pi \times \mathrm{id} \colon P \times [0,1] \to X \times [0,1],$$

where $\pi \colon P \to X$ is the bundle projection (this is just the pullback bundle of P under the projection $X \times [0,1] \to X$). Then F_0 and F_1 define a morphism of G-principal bundles

$$(F_0, F_1) \colon P \times \{0,1\} = (P \times [0,1])|_{X \times \{0,1\}} \to EG,$$

which by Theorem 1.11 can be extended to $P \times [0,1]$, thus yielding a G-homotopy between F_0 and F_1. ☐

Of course for this theory to be of any use we need such universal bundles to exist. The following general existence theorem is due to Milnor.

Theorem 1.12 (OG Milnor paper, Ebert)**.** *For any topological group G there exists a universal G-principal bundle $EG \to BG$, where BG is a CW-complex.*

A priori the space BG constructed in [OG Milnor paper, Ebert] may not be a CW complex. However, if we choose a CW-approximation $f \colon X \to BG$ the total space of the bundle $f^*EG \to X$ is still weakly contractible. This follows by applying the five lemma to the following diagram.

$$\begin{tikzcd}
\pi_{k+1}(X) \arrow[r] \arrow[d, "\cong"] & \pi_k(G) \arrow[r] \arrow[d, "\mathrm{id}"] & \pi_k(f^*EG) \arrow[r] \arrow[d] & \pi_k(X) \arrow[r] \arrow[d, "\cong"] & \pi_{k-1}(G) \arrow[d, "\mathrm{id}"] \\
\pi_{k+1}(BG) \arrow[r] & \pi_k(G) \arrow[r] & \pi_k(EG) \arrow[r] & \pi_k(BG) \arrow[r] & \pi_{k-1}(G)
\end{tikzcd}$$

We can now use the classification results for principal bundles to obtain analogues results for general fiber bundles. As discussed in section 1.1 we can generally recover information about arbitrary fiber bundles by changing the fiber.

Corollary 1.13. *Let G be a topological group acting effectively on a space F. Then for any CW-complex X the map*

$$[X, BG] \to \mathcal{F}_{F,G}(X), \;\; [f] \mapsto [f^*(EG \times_G F)]$$

is a natural bijection, where $\mathcal{F}_{F,G}(X)$ denotes the set of isomorphism classes of fiber bundles with fiber F and structure group G over X.

Proof. By 1.10 the map

$$[X, BG] \to \mathcal{P}_G(X), \;\; [f] \mapsto [f^*EG]$$

is a natural bijection and by Lemma 1.5

$$\mathcal{P}_G(X) \to \mathcal{F}_{F,G}(X), \;\; [P] \mapsto [P \times_G F]$$

is a natural bijection. Composing both thus yields the natural bijection

$$[X, BG] \to \mathcal{F}_{F,G}(X), \;\; [f] \mapsto [(f^*EG) \times_G F]\,.$$

So the statement follows by again applying Lemma 1.5. ☐

Again let us discuss a few examples.

Example 1.14.

(i) The universal cover $\mathbb{R} \to S^1$ is a $\mathbb{Z}$-prinicpal bundle, $\mathbb{Z}$ acting on itself by addition. Since $\mathbb{R}$ is contractible we obtain $B\mathbb{Z} \cong S^1$ and $E\mathbb{Z} \cong \mathbb{R}$.

(ii) From the long exact sequence of homotopy groups

$$\pi_k(EG) \longrightarrow \pi_k(BG) \longrightarrow \pi_{k-1}(G) \longrightarrow \pi_{k-1}(EG)$$

we conclude that $\pi_k(BG) \cong \pi_{k-1}(G)$ for $k \geq 1$. Moreover, BG is connected since it is the image of the connected space EG under $EG \to BG$.

(iii) The universal $O(k)$-prinicpal bundle $EO(k) \to BO(k)$ also has an explicit description. Consider the Stiefel manifold $V_k^n \subseteq \mathbb{R}^{n\times k}$, which is defined as the compact space of orthonormal k-tuples of vectors in $\mathbb{R}^n$. Moreover let Gr_k^n be the Grassmanian of k-planes in $\mathbb{R}^n$, i.e. the set of k-dimensional linear subspaces of $\mathbb{R}^n$. The topology on Gr_k^n is induced by the projection map

$$p\colon V_k^n \to \mathrm{Gr}_k^n,\ (v_1,\dots,v_k) \mapsto \mathrm{span}(v_1,\dots,v_k)$$

Identifying a linear subspace $V \in \mathrm{Gr}_k^n$ with $\mathbb{R}^k$ yields an identification of the fiber $p^{-1}(V)$ with $O(k)$. Moreover the identifications $V \cong \mathbb{R}^k$ can be chosen consistently in a small neighbourhood around V, providing local trivialisations for p. The transition of two such trivialisations is realized by actions of $O(k)$, hence p defines an $O(k)$-principal bundle. Now the canonical inclusion $\mathbb{R}^n \to \mathbb{R}^{n+1}$ induces inclusions $V_k^n \to V_k^{n+1}$ and $\mathrm{Gr}_k^n \to \mathrm{Gr}_k^{n+1}$. Consider the colimits over these inclusions

$$V_k^\infty := \operatorname*{colim}_n (V_k^n), \quad \mathrm{Gr}_k^\infty := \operatorname*{colim}_n (\mathrm{Gr}_k^n)$$

and the induced map $V_k^\infty \to \mathrm{Gr}_k^\infty$. One can choose trivialisations of $V_k^n \to \mathrm{Gr}_k^n$ for varying n, which fit together to induce a trivialisation in the colimit, such that $V_k^\infty \to \mathrm{Gr}_k^\infty$ defines an $O(k)$-principal bundle. We claim that V_k^∞ is weakly contractible. Consider the map

$$V_k^n \to S^{n-1},\ (v_1,\dots,v_k) \mapsto v_1.$$

Similiar arguments as above shows that this map is locally trivial with fiber V_{k-1}^{n-1}. Hence we can consider the long exact sequence of homotopy groups

$$\pi_{i+1}(S^{n-1}) \longrightarrow \pi_i(V_{k-1}^{n-1}) \longrightarrow \pi_i(V_n^k) \longrightarrow \pi_i(S^{n-1})$$

So $\pi_i(V_k^n) \cong \pi_i(V_{k-1}^{n-1})$ for $i \leq n-3$. Applying this inductively yields

$$\pi_i(V_k^n) \cong \pi_i(V_1^{n-(k-1)}) = \pi_i(S^{n-k+1}) = 0$$

for $i \leq n-k-1$. Now from compactness of S^i and the fact that V_k^n is closed in V_k^{n+1} we conclude that the image of any map $S^i \to V_k^\infty$ is contained in some finite stage V_k^n. Therefore $\pi_i(V_k^\infty) = 0$ for all $i \geq 0$, such that the universal $O(k)$-principal bundle is explicitly given by $V_k^\infty \to \mathrm{Gr}_k^\infty$.

Similiarly we get an explicit description of the universal $SO(k)$-prinicpal bundle. Let $\tilde{\mathrm{Gr}}_k^n$ be the set of tuples consisting of a k-dimensional linear subspace of $\mathbb{R}^n$ together with a chosen orientation. The topology on $\tilde{\mathrm{Gr}}_k^n$ is induced by

$$p\colon V_k^n \to \tilde{\mathrm{Gr}}_k^n,\ (v_1,\dots,v_k) \mapsto (\mathrm{span}(v_1,\dots,v_k), \mathrm{o}_{(v_1,\dots,v_k)}),$$

where $\mathrm{o}_{(v_1,\dots,v_k)}$ is the unique orientation on $\mathrm{span}(v_1,\dots,v_k)$, such that $(v_1,\dots,v_k)$ is positively oriented. Similiar to above we see that p defines an $SO(k)$-principal bundle. Taking the colimit, we obtain the universal $SO(k)$-principal bundle $V_k^\infty \to \tilde{\mathrm{Gr}}_k^\infty$.

(iv) Any real k-dimensional vector bundle $E \to X$ can be considered as a fiber bundle with fiber $\mathbb{R}^k$ and structure group $O(k)$. So Corollary 1.13 and the example above show that every such E is the pullback of the universal real k-dimensional vector bundle $V_k^\infty \times_{O(k)} \mathbb{R}^k$ under some map $X \to \mathrm{Gr}_k^\infty$. We show that this universal vector bundle also has a more explicit description, namely as the tautological bundle $\gamma_k \to \mathrm{Gr}_k^\infty$. Recall that this bundle is defined as

$$\gamma_k := \{(X,v) \in \mathrm{Gr}_k^\infty \times \mathbb{R}^\infty \mid v \in X\}$$

together with the canonical projection map to Gr_k^∞. An element in V_k^∞ is a k-tuple of orthonormal vectors in some $\mathbb{R}^n$, which we will subsequently regard as the columns of an $n \times k$ matrix. Then the map

$$V_k^\infty \times \mathbb{R}^k \to \gamma_k, \;\; (A, v) \mapsto (\mathrm{im}(A), Av)$$

descends to the quotient, defining a vector bundle map $V_k^\infty \times_{O(k)} \mathbb{R}^k \to \gamma_k$. This map is clearly a linear isomorphism on fibers, hence an isomorphism of vector bundles.

Similiarly, oriented real k-dimensional vector bundles are classified by maps to $\tilde{\mathrm{Gr}}_k^\infty$. We claim that the universal oriented real k-dimensional vector bundle is given by the tautological bundle $\gamma_k^{SO} \to \tilde{\mathrm{Gr}}_k^\infty$ defined by

$$\gamma_k^{SO} := \{(X, \mathrm{o}_X, v) \in \tilde{\mathrm{Gr}}_k^\infty \times \mathbb{R}^k \mid v \in X\}$$

together with the canonical projection map to $\tilde{\mathrm{Gr}}_k^\infty$. To this end consider the map

$$V_k^\infty \times \mathbb{R}^k \to \gamma_k^{SO}, \;\; (A, v) \mapsto (\mathrm{im}(A), \mathrm{o}_A, Av),$$

where o_A is the unique orientation on $\mathrm{im}(A)$, such that $A\colon \mathbb{R}^k \to \mathrm{im}(A)$ is orientation preserving. Again this map descends to the quotient defining an isomorphism $V_k^\infty \times_{SO(k)} \mathbb{R}^k \to \gamma_k^{SO}$.

(v) Completely analogous to the real case we can also define the complex Stiefel manifolds $V_k^\infty(\mathbb{C})$ and complex Grassmanians $\mathrm{Gr}_k^\infty(\mathbb{C})$ and show that the universal $U(k)$-prinicpal bundle is given by $V_k^\infty(\mathbb{C}) \to \mathrm{Gr}_k^\infty(\mathbb{C})$. The universal complex k-dimensional vector bundle is the tautological bundle $\tau_k \to \mathrm{Gr}_k^\infty(\mathbb{C})$.

(vi) As special cases of the preceeding examples we obtain

$$B\mathbb{Z}_2 \cong \mathbb{R}P^\infty, \quad BS^1 \cong \mathbb{C}P^\infty.$$

1.3 Stiefel-Whitney and Chern-Classes

In the preceeding Chapter we discussed the question of how bundles can be classified. Another fundamental question is of course how to determine, whether two given bundles are isomorphic. As usual in topology, a partial solution to this problem is provided by defining computable invariants. For vector bundles the most important invariants are so called *characteristic classes*. The general definiton of a characteritsic class is as follows. Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$ and write $\mathrm{Vect}_{\mathbb{K}}(B)$ for the set of isomorphism classes of $\mathbb{K}$-vector bundles over a space B. Then $\mathrm{Vect}_{\mathbb{K}}(-)$ defines a contravariant functor $\mathsf{Top} \to \mathsf{Set}$. A characteristic class c for $\mathbb{K}$-vector bundles with values in the cohomology theory H^* is a natural transformation from the functor $\mathrm{Vect}_{\mathbb{K}}(-)$ to the functor H^* regarded as a functor with values in Set. So in more concrete terms c assigns to each $\mathbb{K}$-vector bundle $E \to B$ a cohomology class $c(E) \in H^*(B)$ depending only on the isomorphism class of E, such that for any continuous map $f\colon B' \to B$ the following naturality condition holds

$$c(f^*E) = f^*c(E).$$

Therefore, using the theory of classifiying spaces developed above, we see that for a real (resp. complex) k-dimensional vector bundles, the characteristic class c is uniquely determined by its value for the universal bundle γ_k (resp. τ_k) discussed in example 1.14 (iv) (resp. **??**).

For real vector bundles the most important characteristic classes are the *Stiefel-Whitney classes* and for complex vector bundles the *Chern classes*. Both are formally quite similiar and can be characterized by a few axioms.

Definition 1.15 (Stiefel-Whitney Classes, HATCHER)**.** There is a unique sequence of characteristic classes $w_1, w_2, \ldots$ for real vector bundles with values in $H^*(-; \mathbb{Z}_2)$, such that

i) $w_i(E) \in H^i(B; \mathbb{Z}_2)$ for any real bundle $E \to B$,

ii) $w_i(E) = 0$ for $i > \mathrm{rank}(E)$,

iii) $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$, where $w = 1 + w_1 + w_2 + \ldots$,

iv) For the universal real line bundle $\gamma_1 \to \mathbb{R}P^\infty$, $w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}_2) \cong \mathbb{Z}_2$.

Note that by property ii) the sum $w = 1 + w_1 + w_2 + \dots$ is always finite. The class $w(E)$ is called the *total Stiefel-Whitney class of E*. We sometimes write $w_0(E) := 1 \in H^0(B;\mathbb{Z}_2)$ to simplify notation.

Definition 1.16 (Chern Classes, HATCHER)**.** There is a unique sequence of characteristic classes $c_1, c_2, \dots$ for complex vector bundles with values in $H^*(-;\mathbb{Z})$, such that

i) $c_i(E) \in H^{2i}(B;\mathbb{Z})$ for any complex bundle $E \to B$,

ii) $c_i(E) = 0$ for $i > \operatorname{rank}(E)$,

iii) $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$, where $c = 1 + c_1 + c_2 + \dots$,

iv) For the universal complex line bundle $\tau_1 \to \mathbb{C}P^\infty$, $c_1(\tau_1)$ is a generator of $H^2(\mathbb{C}P^\infty;\mathbb{Z}) \cong \mathbb{Z}$ specified in advance.

Again the *total Chern class* $c = 1 + c_1 + c_2 + \dots$ is well-defined and we write $c_0(E) := 1 \in H^0(B;\mathbb{Z})$. We refer the reader to [HATCHER] for a proof of existence and uniqueness of the Stiefel-Whitney and Chern classes.

Not only are the preceeding definitions formally quite similiar, but there is also a direct connection between the two.

Proposition 1.17 (Hatcher Proposition 3.8)**.** *Regarding a complex rank n vector bundle $E \to B$ as a real rank $2n$ vector bundle, $w_{2i+1}(E) = 0$ and $w_{2i}(E)$ is the image of $c_i(E)$ under the coefficient homomorphism $H^{2i}(B;\mathbb{Z}) \to H^{2i}(B;\mathbb{Z}_2)$.*

Example 1.18.

i) Let $\iota\colon \mathbb{R}P^n \to \mathbb{R}P^\infty$ be the canonical inclusion. Then the pullback $\iota^*\gamma_1$ is isomorphic to the tautological line bundle $\gamma_1^n \to \mathbb{R}P^n$. Since ι induces an isomorphism on cohomology we conclude from naturality that $w_1(\gamma_1^n) = \iota^* w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^n;\mathbb{Z}_2) \cong \mathbb{Z}_2$. Similiarly, $c_1(\tau_1^n)$ of the tautological complex line bundle $\tau_1^n \to \mathbb{C}P^n$ is a generator of $H^2(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}$.

ii) Let $\varepsilon \to B$ be a trivial real bundle over some space B. Then $\varepsilon \cong f^*\varepsilon$ for a constant map $f\colon B \to B$ and thus $w_i(\varepsilon_n) = f^* w_i(\varepsilon) = 0$ for all $i \geq 1$. Similiarly all Chern classes for complex trivial bundles vanish.

iii) Let $TS^n \to S^n$ be the tangent bundle of the sphere and let $\nu_n \to S^n$ be the normal bundle with respect to the standard embedding $S^n \to \mathbb{R}^{n+1}$. Since ν_n and $TS^n \oplus \nu_n$ are trivial we conclude from the example above and property iii) in Definition 1.15 that

$$1 = w(TS^n \oplus \nu_n) = w(TS^n) \smile w(\nu_n) = w(TS^n) \smile 1 = w(TS^n).$$

Thus $w_i(TS^n) = 0$ for all $i \geq 1$.

In the last example we saw that although all Stiefel-Whitney classes of a bundle vanish, it need not be trivial. However for one-dimensional real (resp. complex) bundles it turns out that the first Stiefel-Whitney (resp. Chern) class already completely determines the bundle:

Theorem 1.19 (Hathcer, Proposition 3.10)**.** *Let X be a CW-complex. Then the maps*

$$w_1\colon \mathrm{Vect}_{\mathbb{R}}(X) \to H^1(X;\mathbb{Z}_2)$$

and

$$c_1\colon \mathrm{Vect}_{\mathbb{C}}(X) \to H^2(X;\mathbb{Z})$$

are bijections.

While the first Stiefel-Whitney class of higher dimensional bundles does not identify their isomorphism type it does characterize another important invariant, namely orientability:

Proposition 1.20 (Hatcher, Prop 3.11)**.** *Let X be a CW-complex. A real vector bundle $E \to X$ is orientable if and only if $w_1(E) = 0$.*

Similiarly vanishing of the second Stiefel Whitney class corresponds to the existence of another structure, namely that of a *spin structure*.

Definition 1.21. (i) For $n \geq 2$ the *spin group* $\mathrm{Spin}(n)$ is defined as the total space of the non-trivial two-sheeted cover $\rho\colon \mathrm{Spin}(n) \to \mathrm{SO}(n)$. If $n \geq 3$ this is the universal cover and in the case $n = 2$ we have $\mathrm{Spin}(2) \cong \mathrm{SO}(2)$. For $n = 1$ we define $\mathrm{Spin}(1) := \mathbb{Z}_2$, the total space of the two-sheeted cover $\rho\colon \mathbb{Z}_2 \to \mathrm{SO}(1)$.

(ii) Let $E \to B$ be an oriented real vector bundle equipped with a fiber metric and let $P_{\mathrm{SO}(n)}$ be its oriented orthonormal frame bundle. A *spin structure* on E is a $\mathrm{Spin}(n)$-principal bundle $P_{\mathrm{Spin}(n)}$ together with a map

$$\phi\colon P_{\mathrm{Spin}(n)} \to P_{\mathrm{SO}(n)}$$

of principal bundles, which is ρ-equivariant (compare Definition 1.2).

(iii) An orientable manifold M is called *spin* if its tangent bundle TM admits a spin structure and is called *totally non-spin* if the universal cover of M is not spin.

Remark 1.22. When $n = 1$, $P_{\mathrm{SO}(1)} \cong X$ and a spin structure is simply a two-sheeted cover of X.

Note that the existence of a spin structure does not depend on the chose orientation or fiber metric on E, since different choices yield isomorphic oriented orthonormal frame bundles. In fact the only obstruction to the existence of a spin structure is the second Stiefel-Whitney class, as mentioned above.

Theorem 1.23 (Lawson,Micheaslon)**.** *Let X be a CW-complex. An orientable real vector bundle $E \to X$ admits a spin structure if and only if $w_2(E) = 0$.*

Example 1.24. (i) All spheres S^n, $n \geq 1$ are spin by Example iii).

(ii) The standard example for non-spin manifolds are all even dimensional complex projective spaces $\mathbb{C}P^{2k}$, see [MilnorStasheff??].

Remark 1.25. Let $E \to X$ be a real vector bundle over a CW-complex. If an orientation is given, E is classified by a map $X \to B\mathrm{SO}(n)$. Similiarly, if a spin structure exists, E is classified by a map $X \to B\mathrm{Spin}(n)$, which can be seen as follows. Let $\phi\colon P_{\mathrm{Spin}(n)} \to P_{\mathrm{SO}(n)}$ be a spin structure on E and let $f\colon X \to B\mathrm{Spin}(n)$ be a classifiying map for $P_{\mathrm{Spin}(n)}$. Define a vector bundle

$$\gamma_n^{\mathrm{Spin}} := E\mathrm{Spin}(n) \times_{\mathrm{Spin}(n)} \mathbb{R}^n$$

using Lemma 1.4, where $\mathrm{Spin}(n)$ acts on $\mathbb{R}^n$ by virtue of the covering map $\rho\colon \mathrm{Spin}(n) \to \mathrm{SO}(n)$ and the canonical action of $\mathrm{SO}(n)$. From Lemma 1.5 we obtain

$$f^*\gamma_n^{\mathrm{Spin}} \cong P_{\mathrm{Spin}(n)} \times_{\mathrm{Spin}(n)} \mathbb{R}^n.$$

Moreover, the spin structure ϕ induces an isomorphism $P_{\mathrm{Spin}(n)} \times_{\mathrm{Spin}(n)} \mathbb{R}^n \to P_{\mathrm{SO}(n)} \times_{\mathrm{SO}(n)} \mathbb{R}^n \cong E$ of vector bundles by

$$[(x, v)] \mapsto [(\phi(x), v)].$$

This map is clearly a linear bijection on fibers, hence an isomorphism of vector bundles.

Recall that a manifold M is orientable if and only if its tangent bundle TM restricted to any embedded loop $S^1 \subseteq M$ is trivial. Or in other words "TM does not contain any Moebius Bands". A similiar characterization can be given for spin manifolds, which is why the property of being spin is sometimes referred to as a higher dimensional analogue of orientability.

Proposition 1.26 (Georg Paper, Proposition 2.6)**.** *Let M be a connected oriented manifold of dimension at least* 5.

i) M is spin if and only if for every embedded surface $S \subseteq M$ the restriction $TM|_S$ of the tangent bundle to S is trivial.

ii) M is totally non-spin if and only if there exists an embedded sphere $S^2 \subseteq M$, such that the restriction $TM|_{S^2}$ of the tangent bundle to this sphere is non-trivial.

Proof. Since we will not be making use of the first characterization, we shall only proof part ii) of the proposition and refer the reader to [Georg Paper] for a proof of part i). Let $\tilde{M}$ be the universal cover of M. Since $T\tilde{M}$ does not admit a spin structure we conclude from Theorem 1.23 that $w_2(T\tilde{M}) \neq 0$. So by the universal coefficient theorem there exists a homology class $a \in H_2(\tilde{M};\mathbb{Z}_2)$ with

$$\langle w_2(T\tilde{M}), a\rangle \neq 0.$$

Once more applying the universal coefficient theorem as well as the Hurewicz theorem we obtain an isomorphism

$$\pi_2(\tilde{M}) \otimes \mathbb{Z}_2 \to H_2(\tilde{M};\mathbb{Z}) \otimes \mathbb{Z}_2 \cong H_2(\tilde{M};\mathbb{Z}_2).$$

Explicitly, for some class $\alpha \in \pi_2(\tilde{M})$ represented by a map $f\colon S^2 \to \tilde{M}$ the above isomorphism maps $\alpha \otimes 1$ to $f_*[S^2]_{\mathbb{Z}_2}$, where $[S^2]_{\mathbb{Z}_2}$ is the $\mathbb{Z}_2$-fundamental class of S^2. Therefore we can write $a = f_*[S^2]_{\mathbb{Z}_2}$ for some $f\colon S^2 \to \tilde{M}$. Let $p\colon \tilde{M} \to M$ be the projection and define $g := p\circ f\colon S^2 \to M$. We compute

$$\langle w_2(g^*TM), [S^2]_{\mathbb{Z}_2}\rangle = \langle f^* w_2(p^*TM), [S^2]_{\mathbb{Z}_2}\rangle = \langle w_2(T\tilde{M}), f_*[S^2]_{\mathbb{Z}_2}\rangle = \langle w_2(T\tilde{M}), a\rangle \neq 0,$$

where we used $p^*TM \cong T\tilde{M}$ and naturality of the Kronecker pairing in the second equation. As $\dim(M) \geq 5$ we may assume g to be an embedding and since $w_2(g^*TM) \neq 0$ by the above computation we conclude that g^*TM can not be trivial. □

Finally we record another scenario in which the vanishing of all Stiefel-Whitney classes of a bundle is sufficient for triviality.

Lemma 1.27. *Let $E \to S$ be a real vector bundle of rank at least 3 over a two dimensional CW-complex S. Then E is trivial if and only if $w_1(E) = w_2(E) = 0$.*

Proof. Combining Theorem 1.23 and Remark 1.25 we conclude that there exists a classifiying map $f\colon S \to B\mathrm{Spin}(n)$ for E, where n is the rank of E. From Example 1.14 (ii) it follows that $\pi_k(B\mathrm{Spin}(n)) = 0$ for $k \geq 2$. Since S is two dimensional the map f is homotopic to a constant ([Hatcher, Lemma 4.6]), therefore $E \cong f^*\xi_{\mathrm{Spin}}$ is trivial. □

2 Positive Scalar Curvature

The purpose of this chapter is twofold. The first half provides a brief introduction to scalar curvature, following [Wal17] and [Car21]. In the second half, we prove two concrete existence results for metrics of positive scalar curvature, which will be used in the construction in Chapter ??.

2.1 A brief introduction

Let M^n be an n-dimensional manifold. Given any Riemannian metric g on M, we can associate to it a smooth function $\mathrm{scal}_g \colon M \to \mathbb{R}$ called the *scalar curvature* of M. In the case where M is an embedded surface in $\mathbb{R}^3$, the scalar curvature equals twice the classical Gaussian curvature $K \colon M \to \mathbb{R}$. So in this case we can intuitively quantify the scalar curvature at a given point.

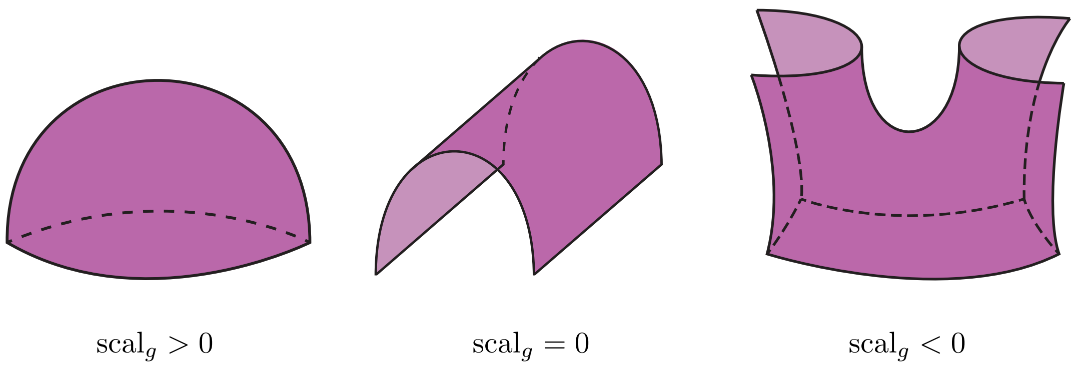


Figure 1: Scalar curvature in dimension 2.

In dimension 2 the scalar curvature is the only intrinsic notion of curvature, however in dimensions $n \geq 3$ there exists a whole hierarchy of different curvatures. The most prominent are sectional curvature, Ricci curvature and scalar curvature, where the sectional curvature is the strongest notion, determining the entire geometry of M. In this thesis we are concerned with the scalar curvature, which is the weakest of the three. In most textbooks on Riemannian geometry it is defined as twice the trace of the Riemannian curvature tensor. While this definiton is best suited for explicit computations it provides no further intuition on what scalar curvature is. Since we have no need for doing such explicit computations, we give a more geometric definition, which essentially extends our intuition in the 2-dimensional case to arbitrary dimensions.

Definition 2.1. Let (M^n, g) be an n-dimensional Riemannian manifold. The scalar curvature $\mathrm{scal}_g(x) \in \mathbb{R}$ at the point $x \in M$ is defined as the value appearing in the third coefficient of the Taylor expansion

$$\frac{\mathrm{vol}(B_r(x))}{\mathrm{vol}(B_r^{\mathrm{eucl.}}(0))} = 1 - \frac{\mathrm{scal}_g(x)}{6(n+2)} r^2 + \dots,$$

where $\mathrm{vol}(B_r(x))$ denotes the volume of a geodesic ball in M of radius r centered at x and $\mathrm{vol}(B_r^{\mathrm{eucl.}}(0))$ denotes the volume of a ball of the same radius r in standard euclidian space $\mathbb{R}^n$.

So intuitively $\mathrm{scal}_g(x) > 0$ means that the volume of balls in M around x of sufficiently small radius r is smaller than the volume of the euclidian ball of the same radius. Similiarly, if $\mathrm{scal}_g(x) < 0$ the volume of the ball in M is larger and if $\mathrm{scal}_g(x) = 0$ the volumes are (asymptotically) identical. Note that this is in accordance with Figure 1.

Example 2.2. (i) We immediately obtain from the definiton that the scalar curvature of $\mathbb{R}^n$ with the usual euclidian metric is zero.

(ii) If (M, g) is one-dimensional, the volume of a radius r ball is simply $2r$. Therefore $\mathrm{scal}_g = 0$.

(iii) The scalar curvature of the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$ can be computed to equal $n(n-1)$.

(iv) Let (M_1, g_1) and (M_2, g_2) be Riemannian manifolds. Then for the scalar curvature of the product $(M_1 \times M_2, g_1 + g_2)$ equipped with the product metric $g_1 + g_2$ one obtains

$\mathrm{scal}_{g_1+g_2}(x_1, x_2) = \mathrm{scal}_{g_1}(x_1) + \mathrm{scal}_{g_2}(x_2)$ for $x_i \in M_i$. We refer the reader to any classical textbook on Riemannian geometry for a proof of these last two examples.

Broadly speaking, the study of scalar curvature falls into two categories: *Rigidity* results and *flexibility* results. As the title of this chapter suggests, one is primarily concerned with results about *positive* scalar curvature. The reason for this is the following theorem.

Theorem 2.3. *Let M^n be a closed manifold of dimension $n \geq 3$ and let $f\colon M \to \mathbb{R}$ be a smooth function.*

(i) If there exists a point $p \in M$ with $f(p) < 0$, then M admits a metric g with scalar curvature $\mathrm{scal}_g = f$.

(ii) If M admits a metric whose scalar curvature is non-negative but not identically zero then M admits another metric g with scalar curvature $\mathrm{scal}_g = f$.

So part (i) of the theorem asserts that the only scalar curvature condition from which we can expect any kind of obstruction results is non-negativity. Moreover, aside from the scalar-flat case (where the scalar curvature is identically zero), part (ii) shows that studying non-negative scalar curvature yields nothing new beyond what is already captured by positive, scalar curvature. In the following we refer to a metric g on M, whose scalar curvature is everywhere positive as a PSC-metric.

The reason why the study of positive scalar curvature has attracted particular attention compared to other curvature conditions, such as positive Ricci or sectional curvature, is that it strikes an exceptionally good balance between the rigidity and flexibility aspects. While positive Ricci or sectional curvature have substantial topological implitcatons, this also makes it significantly more difficult to construct metrics with these curvature conditons. For example the Bonnet-Myers theorem states that a positive lower bound on the Ricci curvature of M already implies that M must be compact with finite fundamental group. For PSC-metrics however, there are existence results under quite general assumptions, which we will discuss below. First let us consider some examples illustrating the rigidity aspects of positive scalar curvature. The most prominent obstruction to the existence of a PSC-metric comes from spin geometry. Let (M^n, g) be a closed Riemannian spin manifold with n even. Then the spin structure and the metric g determine a Hermitian vector bundle $S \to M$ together with a compatible connection ∇^S and a self-adjoint first order differential operator $D\colon \Gamma(S) \to \Gamma(S)$ acting on sections of S. Moreover, there exists a canonical splitting $S \cong S^+ \oplus S^-$ with respect to which D splits as

$$D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}.$$

We call $S \to M$ the *spinor bundle* of M and D the *Dirac operator*. One of the central motivations behind constructing this operator was to find a first order differential operator squaring to the Laplacian. The spinor bundle S carries additional structure making it possible to define such an operator, however, its square agrees with the Laplacian only up to an error term determined by the curvature of M. More precisely, the *Schrödinger–Lichnerowicz formula* holds:

$$D^2 = \Delta^S + \frac{1}{4}\mathrm{scal}_g,$$

where $\Delta^S\colon \Gamma(S) \to \Gamma(S)$ is the connection Laplacian determined by ∇^S. Given any vector bundle $E \to N$ over a closed Riemannian manifold N carrying a fiber metric and a compatible connection, it is a general fact that the induced connection Laplacian is a non-negative operator with respect to the L^2-inner product $\langle \cdot, \cdot \rangle$ on $\Gamma(E)$ induced by the fiber metric and the volume form of N. Applying this to S, let $\psi \in \Gamma(S)$. Then

$$0 \leq \langle D\psi, D\psi \rangle = \langle D^2\psi, \psi \rangle = \langle \Delta^S\psi, \psi \rangle + \frac{1}{4}\int_M \mathrm{scal}_g \|\psi\|^2,$$

where we used self-adjointness of D in the first equation and the Schrödinger–Lichnerowicz formula in the second. If $\mathrm{scal}_g > 0$ and $\psi \neq 0$, the right-hand side is strictly positive, since Δ^S is a non-negative operator. But the right-hand side equals $\langle D\psi, D\psi \rangle = \|D\psi\|^2$, so this forces $D\psi \neq 0$. Hence the kernel of D must be trivial. In particular the integer

$$\mathrm{ind}(D) := \dim(\ker D^+) - \dim(\ker D^-),$$

known as the *index* of D, must vanish (one can show that the kernels of $D^{\pm}$ are finite-dimensional, so this is well defined). Now the major insight is that this index can also be computed in terms of a purely topological invariant of M, namely the so-called $\hat{A}$-class $\hat{A}(TM) \in H^{4*}(M;\mathbb{Q})$ of the tangent bundle $TM \to M$, which is a certain characteristic class. The connection between the index of the Dirac operator and the $\hat{A}$-class of the tangent bundle is established by the famous *Atiyah–Singer index theorem*, which states:

$$\operatorname{ind}(D) = \langle \hat{A}(TM), [M] \rangle,$$

where the right-hand side is the evaluation of the cohomology class $\hat{A}(TM)$ against the fundamental class $[M]$ of M (for the components of $\hat{A}(TM)$ in degree $\neq n$ this is 0, so in particular $\hat{A}(M) = 0$ if n is not divisible by 4). The number $\hat{A}(M) := \langle \hat{A}(TM), [M] \rangle \in \mathbb{Q}$ is also referred to as the $\hat{A}$-*genus* of M. In summary:

Theorem 2.4 (Lichnerowicz, 1963)**.** *Let (M^{4k}, g) be a closed Riemannian spin manifold with positive scalar curvature. Then $\hat{A}(M) = 0$.*

A further, more accessible and geometric example for a rigidity result concerning positive scalar curvature is the following classical result by Llarull.

Theorem 2.5 (Llarull, 1998)**.** *Let M^n be a closed connected Riemannian spin manifold of dimension $n \geq 2$. Suppose $\mathrm{scal}_g \geq n(n-1)$ and let $f\colon M \to S^n$ be a smooth 1-Lipschitz map with $\deg(f) \neq 0$. Then f is an isometry.*

The constant $n(n-1)$ appearing is no coincidence as the scalar curvature of the n-sphere S^n with the standard metric g_0 can be computed to precisely equal this constant. Llarull then applied this theorem in the special case $M = S^n$ and $f = \mathrm{id}$ to show that the only metric g on S^n with $\mathrm{scal}_g \geq n(n-1)$ and $g \geq g_0$ is the standard metric. Said differently, one cannot increase the curvature of the sphere without contracting distances. We conclude our discussion of scalar curvature rigidity results with another quite geometric statement, which is an example for a so called *band width estimate*.

Theorem 2.6 (Gromov)**.** *Let $M = T^{n-1} \times [-1,1]$ and let g be a Riemannian metric on M with scalar curvature bounded from below by a positive constant $\sigma > 0$. Then the distance between the boundary components $\partial_- M = T^{n-1} \times \{-1\}$ and $\partial_+ M = T^{n-1} \times \{1\}$ satisfies*

$$\mathrm{dist}_g(\partial_- M, \partial_+ M) \leq 2\pi \sqrt{\frac{n-1}{n\sigma}}.$$

Intuitively we think of M as a cylinder (a 'band') over T^{n-1} with a left boundary component $\partial_- M$ and a right boundary component $\partial_+ M$. Then this theorem says, that the boundaries of a band, which is highly positively curved must be close together. There are various generalisations of Theorem 2.6 to allow for bands over other manifolds [??]. Gromov even conjectures that T^{n-1} can be replaced by any manifold not admitting a PSC-metric [Conjecture C in Metric inequalities].

Let us now turn to the flexibility aspect of positive scalar curvature. The primary method for constructing PSC-metrics is the following theorem, known as the *Surgery Theorem*, which has been shown independently by Schoen and Yau [SY79] as well as Gromov and Lawson [GL80]. We refer the reader to the Appendix for a brief introduction to surgery and cobordism theory.

Theorem 2.7 (Gromov,Lawson,1980, Shoen Yau 1979, Chernysh Theorem Georg Paper)**.** *Let M be a compact manifold, which admits a* PSC*-metric. Then any manifold M' obtained from M by surgery of codimension at least* 3 *also admits a* PSC*-metric.*

We remark that the construction of the PSC-metric on M' only changes the given PSC-metric on M locally near the embedding along which the surgery is performed. In order to apply this theorem we need a way to control the dimension of the surgeries performed or equivalently the indices of handles in a given cobordism. This question is completely independent of scalar curvature and has for example been studied quite thoroughly in the context of the h-cobordims theorem. One central tool to control the indices of the handles in a cobordism is the following theorem due to Wall

Theorem 2.8 (Wall, 1971)**.** *Let W^{n+1} be a cobordism from M_0 to M_1, such that the inclusion is r-connected for $r \leq \dim(W) - 4$. Then W contains only handles of index $< \dim(W) - r$.*

Combining the above theorem with the Surgery Theorem 2.7, we can make the following observation. If M_0 admits a PSC-metric and the inclusion $M_1 \to W$ is 2-connected, then M_1 also admits a PSC-metric. In fact, using the conclusion of Theorem 2.8 it is possible to obtain an even stronger result. We say that a metric g on a manifold M with boundary is of *product form near the boundary*, if there is some collar neighbourhood $\iota\colon \partial M \times [0,1) \to M$, such that $\iota^* g = g|_{\partial M} + dt^2$, where $g|_{\partial M}$ is the induced metric on ∂M.

Theorem 2.9 (extending psc to corbodism, Georg,Gaja)**.** *Let W^{n+1} be a cobordism from M_0 to M_1, such that the inlcusion $M_1 \to W$ is 2-connected. Then any PSC-metric on M_0 can be extended to a PSC-metric on W, which is of product form near the boundary.*

In view of the above discussion, one strategy to construct a psc metric on a given closed manifold M_1 of dimension at least 5 is to first find a cobordism W from some psc manifold M_0 to M_1 and then modify this cobordism by surgeries to obtain another cobordism W' from M_0 to M_1, such that the inclusion $M_1 \to W'$ is 2-connected. Of course, such a modification by surgeries may not always be possible, so our next goal is to explain certain scenarios in which it is.

2.2 Existence results for PSC-metrics

We will proof the following two existence results for PSC-metrics, one in the spin case and one in the totally non-spin case.

Theorem 2.10. *Let W^{n+1} be a cobordism from M_0 to M_1, $n \geq 5$, such that W is spin. Suppose that M_1 is connected, the inlcusion $M_1 \to W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W, which is of product form near the boundary.*

Theorem 2.11. *Let W^{n+1} be an oriented cobordism from M_0 to M_1, $n \geq 5$. Suppose that M_1 is a connected totally non-spin manifold, the inlcusion $M_1 \to W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W, which is of product form near the boundary.*

We first show the totally non-spin case.

Proof of Theorem 2.11. Without loss of generality assume W to be connected as well (otherwise only consider the connected component containing M). We first show that $\pi_2(W, M)$ is finitely generated. Let $\tilde{M}$ and $\tilde{W}$ be the universal covers of M and W respectively. Then the inclusion $M \to W$ lifts to a map $\tilde{M} \to \tilde{W}$ by virtue of which we can consider the relative homotopy group $\pi_2(\tilde{W}, \tilde{M})$. It follows from the Hurewicz Theorem that

$$\pi_2(\tilde{W}, \tilde{M}) \cong H_2(\tilde{W}, \tilde{M}).$$

By ??? $H_2(\tilde{W}, \tilde{M})$ is finitely generated as a $\mathbb{Z}[\pi_1(M)]$ module (??) hence the same holds for $\pi_2(\tilde{W}, \tilde{M})$. Now consider the following diagram where the rows are exact.

$$\begin{tikzcd}
\pi_2(\tilde{M}) \arrow[r] \arrow[d, "\cong"] & \pi_2(\tilde{W}) \arrow[r, two heads] \arrow[d, "\cong"] & \pi_2(\tilde{W}, \tilde{M}) \arrow[d] \\
\pi_2(M) \arrow[r] & \pi_2(W) \arrow[r, two heads] & \pi_2(W, M)
\end{tikzcd}$$

The two rightmost horizontal maps are surjective by exactness, since $\pi_1(\tilde{M}) = 0$ and $\pi_1(M) \to \pi_1(W)$ is an isomorphism. A diagram chase now shows that $\pi_2(\tilde{W}, \tilde{M}) \to \pi_2(W, M)$ is an isomorphism.

Next find lifts with... Let

$$F\colon W \to B := B\mathrm{SO}(n+1)$$

be a classifying map for the tangent bundle TW and set $f := F \circ i$, where $i\colon M_1 \to W$ is the inlcusion. Then f is a classifying map for $TM_1 \oplus \underline{\mathbb{R}}$ with $\underline{\mathbb{R}}$ the trivial line bundle. We claim that f is surjective on π_2. Since $\pi_2(B) \cong \pi_1(\mathrm{SO}(n+1)) \cong \mathbb{Z}_2$ it suffices to show that $\pi_2(f)$ is non-zero. By Proposition 1.26 there exists an embedding $\iota\colon S^2 \to M_1$, such that $w_2(\iota^* TM_1) \neq 0$. We compute

$$w_2((f \circ \iota)^* \gamma_n^{\mathrm{SO}}) = w_2(\iota^*(TM_1 \oplus \underline{\mathbb{R}})) = w_2(\iota^* TM_1) \neq 0.$$

Therefore $f \circ \iota$ represents a non-trivial element in $\pi_2(B)$ showing the claim. Now consider the following diagram, where the row is exact.

$$\begin{tikzcd}
\pi_2(M_1) \arrow[r, "i_*"] \arrow[dr, "f_*"'] & \pi_2(W) \arrow[r] \arrow[d, "F_*"] & \pi_2(W, M_1) \\
& \pi_2(B) &
\end{tikzcd}$$

Using surjectivity of f_*, a diagram chase shows that we can find lifts $\alpha_1, \ldots, \alpha_l \in \pi_2(W)$ of the generators $\alpha_1^{\mathrm{rel}}, \ldots, \alpha_l^{\mathrm{rel}} \in \pi_2(W, M_1)$, which lie in the kernel of F_*. Moreover, since $n \geq 5$ the classes α_i can be represented by embeddings

$$\phi_i \colon S^2 \to W.$$

We show that the normal bundle ν_i of ϕ_i is trivial. Since W is oriented it follows from Proposition 1.20 that $w_1(\phi_i^* TW) = 0$. Furthermore, $F \circ \phi_i$ is null-homotopic by construction, such that

$$w_2(\phi_i^* TW) = w_2((F \circ \phi_i)^* \gamma_n^{\mathrm{SO}}) = 0.$$

Using $w_k(TS^2) = 0$ (see Example 1.18 iii)) we compute

$$w_k(\nu_i) = w_k(\nu_i \oplus TS^2) = w_k(\phi_i^* TW) = 0$$

for $k = 1, 2$. Thus Lemma 1.27 shows that ν_i is indeed trivial. This allows us to perform surgery to obtain a new cobordism W' from M_0 to M_1 with

$$\pi_2(W', M_1) = 0$$

and $\pi_1(W', M_1) \cong \pi_1(W, M_1) = 0$, see Proposition A.2 (ii). In other words, the inclusion $M_1 \to W'$ is 2-connected. So by Theorem 2.9 there exists a psc metric g on W', which is of product form near the boundary. To construct a psc metric on the original cobordism W, observe that since W' was obtained from W by surgeries of dimension 2, W can be obtained from W' by surgeries of codimension 3. Therefore the surgery theorem 2.7 applies to construct a psc metric on W. Since this construction only changes the given metric g locally near the surgery embeddings, the resulting metric on W is still of product form near the boundary. □

Proof of Theorem 2.10. Again $\pi_2(W, M)$ is finitely generated as a $\mathbb{Z}[\pi_1(M)]$-module. Let $\alpha_1^{\mathrm{rel}}, \ldots, \alpha_l^{\mathrm{rel}}$ be such generators. We could argue as in the proof of Theorem 2.11, replacing $B\mathrm{SO}(n+1)$ by $B\mathrm{Spin}(n+1)$ and observing that the corresponding f is trivially surjective on π_2 as $\pi_2(B\mathrm{Spin}(n+1)) \cong \pi_1(\mathrm{Spin}(n+1)) = 0$. However this just amounts to the following more direct argument. Choose lifts $\alpha_1, \ldots, \alpha_l \in \pi_2(W)$ of $\alpha_1^{\mathrm{rel}}, \ldots, \alpha_l^{\mathrm{rel}}$ and represent α_i by an embedding $\phi_i \colon S^2 \to W$. Then

$$w_k(\phi_i^* TW) = \phi_i^* w_k(TW) = 0$$

for $k = 1, 2$ since W is spin. The rest of the proof is now as in Theorem 2.11. □

Remark 2.12. The similarity between the proofs of Theorems 2.10 and 2.11 is no coincidence, since both statements are a consequence of a more general result. This unifying approach can be formulated using the language of tangential 2-types, which we shall not discuss further in this thesis. See for example [Georg Paper etc.].

In the main construction in Chapter 4 we will invoke Theorem 2.11 to construct a PSC-metric, so let us conclude this section by discussing some basic results for identifying totally non-spin manifolds.

Lemma 2.13. *(i) The product of a totally non-spin manifold with any non-empty manifold is totally non-spin.*

(ii) A manifold containing a totally non-spin submanifold of codimension 0 *is totally non-spin.*

(iii) If M is totally non-spin and of dimension $n \geq 3$, then $M - D^n$ is totally non-spin for any embedded disk $D^n \subseteq \mathrm{int}(M)$ in the interior of M.

Remark 2.14. All statements of the above lemma remain valid if "totally non-spin" is replaced by "non-spin". The corresponding proofs are just simpler versions of the arguments given below.

Proof. For (i) let M_1 and M_2 be manifolds with M_1 totally non-spin. Then the universal cover of the product $M_1 \times M_2$ is given by the product $\tilde{M}_1 \times \tilde{M}_2$ of the respective universal covers. Denote by

$$\pi_1 \colon \tilde{M}_1 \times \tilde{M}_2 \to \tilde{M}_1, \ \ \pi_2 \colon \tilde{M}_1 \times \tilde{M}_2 \to \tilde{M}_2$$

the projections and let $\iota \colon \tilde{M}_1 \to \tilde{M}_1 \times \tilde{M}_2$ be an inclusion for some choice of basepoint in $\tilde{M}_2$. We compute

$$\begin{aligned} \iota^* w_2(T(\tilde{M}_1 \times \tilde{M}_2)) &= \iota^* w_2(\pi_1^* T\tilde{M}_1 \oplus \pi_2^* T\tilde{M}_2) \\ &= w_2((\pi_1 \circ \iota)^* T\tilde{M}_1 \oplus (\pi_2 \circ \iota)^* T\tilde{M}_2) \\ &= w_2(T\tilde{M}_1) \neq 0, \end{aligned}$$

where we used that $\pi_1 \circ \iota$ is the identity and that $\pi_2 \circ \iota$ is constant, such that $(\pi_2 \circ \iota)^* T\tilde{M}_2$ is trivial. Thus $w_2(T(\tilde{M}_1 \times \tilde{M}_2)) \neq 0$.

For (ii) suppose that $N \subseteq M$ is some totally non-spin codimension 0 submanifold of the manifold M. Let $\iota \colon N \to M$ be the inclusion and consider its lift to the universal covers:

$$\begin{array}{ccc} \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \\ \downarrow & & \downarrow \\ N & \xrightarrow{\iota} & M \end{array}$$

Since $\tilde{\iota}$ is a local diffeomorphism, the differential $d\tilde{\iota} \colon T\tilde{N} \to T\tilde{M}$ is a fiber-wise isomorphism, such that the following is a pullback diagram

$$\begin{array}{ccc} T\tilde{N} & \xrightarrow{d\tilde{\iota}} & T\tilde{M} \\ \downarrow & & \downarrow \\ \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \end{array}$$

Therefore $\tilde{\iota}^* w_2(T\tilde{M}) = w_2(T\tilde{N}) \neq 0$ and so $w_2(T\tilde{M}) \neq 0$.

For (iii) assume without of loss generality that M is connected. Denote by $B^n := \mathrm{int}(D^n)$ the open ball. Then $M - D^n$ is the interior of $M - B^n$ from which we conclude that $M - D^n$ is totally non-spin if and only if $M - B^n$ is totally non-spin. This follows using an analogous argument as in (ii) by noting that for a manifold X with boundary the universal cover of $\mathrm{int}(X)$ is given by $\mathrm{int}(\tilde{X})$ and the inlcusion $\mathrm{int}(\tilde{X}) \to \tilde{X}$ is a homotopy equivalence. We will now show that $M - B^n$ is totally non-spin. Let $p \colon \tilde{M} \to M$ be the universal cover of M and write $M_0 := M - B^n$. We claim that the universal cover of M_0 agrees with the covering

$$\hat{M}_0 = \tilde{M}|_{M_0} = \tilde{M} - p^{-1}(B^n) \xrightarrow{p} M_0.$$

First of all $\hat{M}_0$ is connected as $n \geq 2$. We now have to show that $\hat{M}_0$ is simply connected. Let $F = p^{-1}(x)$ be some fiber of p. From the classification of coverings we see that

$$p^{-1}(B^n) \cong B^n \times F$$

since B^n is simply connected. If the fiber F is finite we can apply the Seifert-van-Kampen Theorem to conclude that $\tilde{M}$ with a single disk B^n removed is still simply connected, such that the statement follows inductively. If the fiber is not finite we can argue more generally as follows. Let $\phi \colon S^1 \to \hat{M}_0$ be some smooth loop. Certainly we can contract this loop in $\tilde{M}$, that is we can find some map $\Phi \colon D^2 \to \tilde{M}$ restricting to α on the boundary. We now show that Φ is homotopic rel S^1 to some map $\Phi_1 \colon D^2 \to \tilde{M}$, such that the image of Φ_1 is already contained in $\hat{M}_0$, from which the claim follows. To construct such a homotopy let $0 \in B^n \subseteq M$ be the center of the embedded disk. After a preliminary homotopy we may assume that Φ is transversal to $p^{-1}(0)$. The crucial observation is that, since $n \geq 3$, this is equivalent to requiring that Φ does not intersect $p^{-1}(0)$. Now note that S^{n-1} is a strong deformation retract of $D^n - 0$ and applying such a deformation retraction on each disk $D^n - 0 \subseteq p^{-1}(D^n - 0)$ shows that $\hat{M}_0$ is a strong deformation retract of $\tilde{M} - p^{-1}(0)$. Let

$$r\colon \tilde{M} - p^{-1}(0) \to \hat{M}_0$$

be such a retraction and let

$$r_t\colon \tilde{M} - p^{-1}(0) \to \tilde{M} - p^{-1}(0)$$

be a homotopy $\operatorname{rel} S^1$ from the identity to r (composed with the corresponding inclusion). Then $\Phi_t := r_t \circ \Phi$ defines a homotopy $\operatorname{rel} S^1$ from Φ to some map Φ_1 whose image is contained in $\hat{M}_0$. This shows the claim and hence the universal cover of M_0 is given by

$$\hat{M}_0 = \tilde{M} - p^{-1}(B^n) \cong \tilde{M} - (B^n \times F).$$

Finally, the Mayer-Vietoris sequence of the pushout

$$\begin{tikzcd} S^{n-1} \times F \arrow[r] \arrow[d] & D^n \times F \arrow[d] \\ \tilde{M} - (B^n \times F) \arrow[r, "\iota"] & \tilde{M} \end{tikzcd}$$

where all maps are inclusions, then shows that ι is injective on $H^2(-;\mathbb{Z}_2)$. Therefore

$$w_2(T\hat{M}_0) = w_2(\iota^* T\tilde{M}) = \iota^* w_2(T\tilde{M}) \neq 0.$$

$\square$

3 Enlargeability

In this section we discuss the concept of enlargeability, which is the central idea the results in this thesis are concerned with.

Definition 3.1. A closed oriented Riemannian n-manifold M is called enlargeable if for all $\varepsilon > 0$ there exists a Riemannian covering $\hat{M} \to M$ together with an ε-contracting (that is ε-Lipschitz) map $\hat{M} \to S^n$, which is constant outside a compact subset and of non-zero degree.

Note that this notion does not depend on the particular choice of Riemannian metric on M, since for closed manifolds all metrics are in bi-Lipschitz correspondence. Intuitively, the covering $\hat{M}$ can be pictured as an "expansion" of M itself (think for example of the usual sketches of coverings of S^1). Enlargeability now means that this expansion can be chosen arbitrarily large, such that it can be wrapped as tightly around a sphere as one wishes. Naively one might expect that for an enlargeable manifold M the universal cover $\tilde{M}$ - which in some sense is the "largest" covering - is hyperspherical, meaning that it admits an ε-contracting map $\tilde{M} \to S^n$, which is constant outside a compact subset and of non-zero degree for arbitrarily small $\varepsilon > 0$. However, Brunnbauer and Hanke have constructed counterexamples to this naive expectation in [BH, Theorem 1.4].

The simplest example of an enlargeable manifold is the Torus T^n, since it is covered by $\mathbb{R}^n$, which is clearly hyperspherical. More generally the same holds for any closed manifold admitting a metric of non-positive sectional curvature by the Cartan-Hadamard Theorem. Non-enlargeable manifolds are for example all simply connected manifolds, since no interesting covers exist in this case.

The central fact about enlargeable manifolds is that they do not admit psc metrics.

Theorem 3.2 (GL1980, GL1982, CS2020)**.** *Enlargeable manifolds do not admit metrics of positive scalar curvature.*

This is the result that motivated the original definition by Gromov and Lawson [GL1980], where they first showed that enlargeability is an obstruction to the existence of psc metrics, under the additional assumption that the coverings $\hat{M}$ in Definition 3.1 are finite an spin. They were later able to drop the finiteness assumption [GL1982]. Both proofs are based on techniques using the spin Dirac operator, so it was unclear how and if these results carry over to the non-spin case. Rather recently however, Cecchini and Schick resolved this issue in [CS2020] by relying on modern minimal surface methods developed by Schoen and Yau [SY...].

The following Lemma now provides two easy ways to construct new enlargeable manifolds out of given ones.

Lemma 3.3. *(i) The product of enlargeable manifolds is enlargeable.*

(ii) A closed oriented manifold admitting a non-zero degree map onto an enlargeable manifold is enlargeable.

Proof. For **??** let M^m and N^n be enlargeable. For $\varepsilon > 0$ choose Riemannian coverings $p\colon \hat{M} \to M$ and $q\colon \hat{N} \to N$ as well as ε-contracting maps $f\colon \hat{M} \to S^m$ and $g\colon \hat{N} \to S^n$ of non-zero degree. Then

$$p \times q\colon \hat{M} \times \hat{N} \to M \times N$$

is a Riemannian covering, where both manifolds are equipped with the corresponding product metric. Moreover

$$f \times g\colon \hat{M} \times \hat{N} \to S^m \times S^n$$

defines an ε-contracting map of degree $\deg(f \times g) = \deg(f)\deg(g) \neq 0$. Now consider the map

$$\phi\colon S^m \times S^n \to S^m \wedge S^n \cong S^{m+n},$$

given by projection onto the smash product

$$S^m \wedge S^n := {S^m \times S^n}/{(S^m \times * \cup * \times S^n)},$$

which is well known to be homeomorphic to the sphere S^{m+n}. The map ϕ is easily seen to be of degree ± 1 (depending on the chosen orientations) and after a slight perturbation we may also assume it to be smooth. Moreover, because $S^m \times S^n$ is closed, it follows that ϕ is a λ-Lipschitz map for some constant $\lambda > 0$. Therefore $\phi \circ (f \times g)$ is a $\lambda\varepsilon$-Lipschitz map of non-zero degree and since ε was chosen arbitrarily the statement follows.

For **??** let M^m be enlargeable and let $h\colon P \to M$ be some non-zero degree map. Again given $\varepsilon > 0$ choose a Riemannian covering $p\colon \hat{M} \to M$ as well as an ε-contracting map $f\colon \hat{M} \to S^m$. Define a covering $\pi\colon \hat{P} \to P$ as the pullback covering of p under h, that is π fits into a pullback diagram of the form

$$\begin{array}{ccc} \hat{P} & \xrightarrow{H} & \hat{M} \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle p} \\ P & \xrightarrow{h} & M \end{array}$$

where H is some suitable smooth map. Since $\hat{M}$ and P are closed, the same holds for $\hat{P}$. Moreover h is μ-Lipschitz for some constant $\mu > 0$, because P is closed. Consequently H is also μ-Lipschitz, given that both π and p are local isometries. For the degree of H we compute

$$\deg(p)\deg(H) = \deg(p \circ H) = \deg(h \circ \pi) = \deg(h)\deg(\pi) \neq 0,$$

using the fact that $\deg(h) \neq 0$ by assumption and that the degree of a covering is given by the cardinality of the fiber. We conclude that H is of non-zero degree, such that $f \circ H\colon \hat{P} \to S^m$ is a $\mu\varepsilon$-Lipschitz map of non-zero degree. Again since ε was chosen arbitrarily the statement follows. □

From Part **??** of the above lemma it follows immediately, that enlargeability is a homotopy invariant, since a homotopy equivalence of connected closed oriented manifolds has degree ± 1. In fact much more is true, namely that enlargeability is a homological invariant in the following sense.

Theorem ([BH10])**.** *Let M be a closed oriented n-manifold and let $\phi\colon M \to B\pi_1(M)$ be the classifying map of the universal cover. Then M is enlargeable if and only if $\phi_*[M] \in H_n(B\pi_1(M); \mathbb{Q})$ does not lie in the rational vector subspace*

$$H_n^{\text{small}}(B\pi_1(M); \mathbb{Q}) \subseteq H_n(B\pi_1(M); \mathbb{Q})$$

of "small" classes, i.e. $\phi_*[M]$ is "large".

In [BH10] the authors showed this by giving a general definition of "small" and "large" homology classes of a simplicial complex X with finitely generated fundamental group, such that the assignment $X \mapsto H_n^{\text{small}}(X; \mathbb{Q})$ is functorial for continuous maps $f\colon X \to Y$ inducing an epimorphism on π_1, that is $f_*\colon H_n(X; \mathbb{Q}) \to H_n(Y; \mathbb{Q})$ maps small classes to small classes. Moreover if f happen to induce an isomorphism on π_1 then f_* also maps large classes to large classes. This general definition then specializes to the fact that a closed oriented manifold M is enlargeable if and only if the fundamental class $[M]$ is large, that is $[M] \notin H_n^{\text{small}}(M; \mathbb{Q})$. Combining these observations, the theorem follows. Furthermore this argument shows that the notion of enlargeability can be purely thought of as a concept of simplicial complexes. For details we refer the reader to [BH10].

4 Main Construction

4.1 Construction of Circle Bundles with PSC

In the beginning we raised the following question.

Question. Does there exist a circle bundle $E \to M$ over an enlargeable manifold M, such that the total space E admits a PSC-metric?

We will now construct examples of such bundles in all dimensions $\dim(M) \geq 4$. Let us first describe the strategy behind the construction. The existence of a PSC-metric on E will be established as an application of Theorem 2.11, restated here for convenience:

Theorem 2.11. *Let W^{n+1} be an oriented cobordism from M_0 to M_1, $n \geq 5$. Suppose that M_1 is a connected totally non-spin manifold, the inlcusion $M_1 \to W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W, which is of product form near the boundary.*

In order to apply this in our case, we make the following two observations, which hold true for any circle bundle $E \to M$ over a closed orientable manifold M.

(i) There exists an oriented nullbordism of E, namely the total space of the associated disk bundle $DE \to M$,

(ii) The inclusion $E \cong \partial DE \to DE$ induces an isomorphism on π_1 if and only if the fibers of E are contractible, that is the inclusion $S^1 \to E$ is trivial on π_1.

We have already observed in Example (iii) that DE is a nullbordism of E. Moreover, the structure group of $DE \to M$ acts on fibers by orientation preserving diffeomorphisms. Thus any orientation on M gives rise to an orientation on DE, which is uniquely determined by requiring local trivialisations to be orientation preserving. For (ii) note that $DE \to M$ is a homotopy equivalence. Hence the diagram

$$\begin{tikzcd} E \arrow[r] \arrow[dr] & DE \arrow[d, "\simeq"] \\ & M \end{tikzcd}$$

shows that $E \to DE$ induces an isomorphism on π_1 if and only if $E \to M$ does. Then the claim follows follows from the long exact sequence of homotopy groups:

$$\pi_1(S^1) \longrightarrow \pi_1(E) \longrightarrow \pi_1(M) \longrightarrow \pi_0(S^1)$$

So to answer the question above positively, we have to construct circle bundles $E \to M$ with the following properties.

(I) M is enlargeable, orientable and connected,

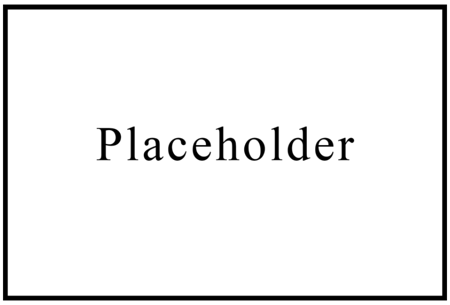
(II) E is totally non-spin,

(III) The inclusion $S^1 \to E$ is trivial on π_1.

We define M as the connected sum of three closed manifolds. Let N^{2n} be any $2n$-dimensional orientable totally non-spin manifold. For example take $N := \mathbb{C}P^2 \times T^{2n-4}$, which is totally non-spin by Lemma 2.13 and Example (ii). Then consider

$$M := \mathbb{C}P^n \# T^{2n} \# N.$$

Intuitively the bundle $E \to M$ will now look as follows. It is trivial over the summands T^{2n} and N, and is given by the canonical bundle $S^{2n+1} \to \mathbb{C}P^n$ over the $\mathbb{C}P^n$ summand (or rather the restriction of this bundle to $\mathbb{C}P^n - D^{2n} \subseteq M$). Schematically:



The presence of the Torus summand then corresponds to property (I), the trivial bundle over N to property (II) and the S^{2n+1} over $\mathbb{C}P^n$ to property (III). We will now construct this bundle rigourously and subsequently discuss the proof of properties (I) - (III) in more detail. Let $\alpha \in H^2(\mathbb{C}P^n)$ be the generator given boundary the first chern class of the tautological complex line bundle $\tau_1^n \to \mathbb{C}P^n$. Implicitly all homology and cohomology groups in this chapter are taken with $\mathbb{Z}$-coefficients. Consider the isomorphism

$$\phi\colon H^2(\mathbb{C}P^n) \oplus H^2(T^{2n}) \oplus H^2(N) \to H^2(M)$$

and let $c_1 \in H^2(M)$ be the class corresponding to $(\alpha, 0, 0)$ under this isomorphism. By Theorem 1.19 there exists a unique complex line bundle $L \to M$ with first chern class $c_1(L) = c_1$. Now define $E \to M$ as the S^1-principal bundle associated to this line bundle. We claim that E is of the form as described above. To this end we will show that $L \to M$ is trivial over the summands T^{2n} and N, and is given by the restriction of $\tau_1^n \to \mathbb{C}P^n$ to $\mathbb{C}P^n - D^{2n} \subseteq M$ over the $\mathbb{C}P^n$ summand. By Example 1.7 (ii) the associated prinicpal bundle of $\tau_1^n \to \mathbb{C}P^n$ is precisely $S^{2n+1} \to \mathbb{C}P^n$, such that the claim then follows from the naturality of changing the fiber, see Remark 1.6. To establish the statements made about $L \to M$ let

$$i\colon N - D^{2n} \to M, \quad i_1\colon N - D^{2n} \to N$$

be the inclusions. Then

$$\begin{tikzcd}
H^2(M) \arrow[rr, "i^*"] & & H^2(N - D^{2n}) \\
H^2(\mathbb{C}P^n) \oplus H^2(T^{2n}) \oplus H^2(N) \arrow[u, "\phi"', "\cong"] \arrow[rr, "\text{proj.}"] & & H^2(N) \arrow[u, "i_1^*"', "\cong"]
\end{tikzcd}$$

commutes and therefore $c_1(i^*L) = i*(c_1) = 0$. Again using Theorem 1.19 we conclude that i^*L is trivial. An analogous argument shows triviality of L over the T^{2n} summand. Similiarly, if we denote by

$$j\colon \mathbb{C}P^n - D^{2n} \to M, \quad j_1\colon \mathbb{C}P^n - D^{2n} \to \mathbb{C}P^n$$

the inclusions, the diagram

$$\begin{tikzcd}
H^2(M) \arrow[rr, "j^*"] & & H^2(\mathbb{C}P^n - D^{2n}) \\
H^2(\mathbb{C}P^n) \oplus H^2(T^{2n}) \oplus H^2(N) \arrow[u, "\phi"', "\cong"] \arrow[rr, "\text{proj.}"] & & H^2(\mathbb{C}P^n) \arrow[u, "j_1^*"', "\cong"]
\end{tikzcd}$$

commutes and thus $c_1(j^*L) = j^*(c_1) = j_1^*(\alpha) = c_1(j_1^*\tau_1^n)$. Hence, applying Theorem 1.19 once more shows that j^*L and $j_1^*\tau_1^n$ are isomorphic bundles. This concludes the construction of E. Let us now verify properties (I) - (III).

Proof of property (I). M is connected and orientable, as each summand is. Moreover, since the collapse map $M \to T^{2n}$ has degree 1 and T^{2n} is enlargeable, it follows from Lemma 3.3 that M is also enlargeable. $\square$

Proof of property (II). Let $i\colon N - D^{2n} \to M$ be the inclusion. We have established above, that

$$i^*E \cong (N - D^{2n}) \times S^1.$$

From Lemma 2.13 (i) and (iii) we conclude that $(N - D^{2n}) \times S^1$ is totally non-spin. Therefore E contains a totally non-spin submanifold of codimension 0 and so Lemma 2.13 (ii) implies that E is also totally non-spin. $\square$

Proof of property (III). Intuitively we can contract the fibers over the S^{2n+1} contained in E. One way to formalize this is the following. Consider the following commutative diagram, where the rows are taken from the long exact sequences of homotopy groups of the respective fibrations and the vertical maps are induced by inclusions:

$$\begin{tikzcd}
\pi_2(M) \arrow[r, "\partial_1"] & \pi_1(S^1) \arrow[r] & \pi_1(E) \\
\pi_2(\mathbb{C}P^n - D^{2n}) \arrow[u] \arrow[r, "\partial_2"] \arrow[d, "\cong"'] & \pi_1(S^1) \arrow[r] \arrow[u, "\mathrm{id}"'] \arrow[d, "\mathrm{id}"'] & \pi_1(j^*E) \arrow[u] \arrow[d] \\
\pi_2(\mathbb{C}P^n) \arrow[r, "\partial_3"] & \pi_1(S^1) \arrow[r] & \pi_1(S^{2n+1})
\end{tikzcd}$$

The vertical map on the lower left is an isomorphism, because both $\mathbb{C}P^n$ and $\mathbb{C}P^n - D^{2n}$ are simply connected and so by the Hurewicz Theorem

$$\begin{tikzcd}
\pi_2(\mathbb{C}P^n - D^{2n}) \arrow[r] \arrow[d, "\cong"] & \pi_2(\mathbb{C}P^n) \arrow[d, "\cong"] \\
H_2(\mathbb{C}P^n - D^{2n}) \arrow[r] & H_2(\mathbb{C}P^n)
\end{tikzcd}$$

commutes and the vertical maps are isomorphisms. Since the bottom horizontal map is an isomorphism the same holds for the top horizontal map. We now claim that ∂_1 is surjective. For this it suffices to show surjectivity of ∂_2, which in turn is equivalent to surjectivity of ∂_3. But this is clear by exactness of the rows and $\pi_1(S^{2n+1}) = 0$. Consequently

$$\pi_1(S^1) \to \pi_1(E)$$

is the zero map. □

The above construction now gives examples for circle bundles with positive scalar curvature over enlargeable manifolds M in all even dimensions $\dim(M) \geq 4$. To also get examples in odd dimensions let $E \to M$ be as above and let $\pi\colon M \times S^1 \to M$ be the projection. Then by Example 2.2 (ii) and (iv) the pullback bundle $\pi^*E \cong E \times S^1$ admits a metric of positive scalar curvature. Furthermore the base space $M \times S^1$ is still enlargeable by Lemma 3.3. Naturally, the question arises as to what happens in dimensions $\dim(M) \leq 3$. In this case it turns out that no circle bundle $E \to M$ over an enlargeable manifold M admits a metric of positive scalar curvature. This was shown in [KS25, Theorem B, Remark 2].

4.2 Necessity of the Enlargeability Assumption in Gromov's Decay Theorem

Let us come back to the Theorem, which originally motivated the whole discussion.

Theorem (Enlargeable Decay Theorem, [Gro18], page 15f.). *Let X be a complete Riemannian manifold, $V \to M$ a two dimensional vector bundle over an enlargeable manifold M. Assume that*

(i) There exists a proper map $f\colon X \to V$, $\deg(f) \neq 0$,

(ii) The total space of the circle bundle corresponding to V is also enlargeable.

Then

$$\min_{B_{x_0}(R)} \mathrm{scal}(X) \leq \frac{8\pi^2}{(R - R_0)^2}$$

for some $R_0 = R_0(X, x_0)$ and all $R > R_0$.

We have seen above that for non-trivial bundles $V \to M$ condition (ii) need not be satisfied. Nonetheless, this does not yet answer the question of whether this condition is even necessary. In this section we will use the preceeding results to construct counterexamples to the Theorem if one drops condition (ii) entirely. Specifically, we construct vector bundles $V \to M$ over enlargeable manifolds M, such that V admits a complete metric of uniformly positive scalar curvature. Then setting $X := V$ and $f := \mathrm{id}$ provides the desired counterexamples.

Let $E \to M$ be a circle bundle as constructed in the previous section. The existence of a PSC-metric on E followed from an application of Theorem 2.11. However, we did not make use of the full strength of this theorem. Not only do we obtain a PSC-metric on E but also a PSC-metric on the nullbordism DE, which is of product form near the boundary. This observation allows for the following construction. Let g be such a PSC-metric on DE and denote the induced metric on E by g_E. Choose some collar neighbourhood $\iota\colon E \times [0, 1) \to DE$ with respect to which g is in product form, i.e.

$$\iota^* g = g_E + dt^2.$$

Moreover, consider the Riemannian manifold $(E \times [1, \infty), g_E + dt^2)$ equipped with the identity as a collar neighbourhood. We can glue this manifold with (DE, g) along the common boundary E to define another Riemannian manifold

$$(V', h) := (E \times [1, \infty) \cup_E DE, (g_E + dt^2) \cup_E g).$$

We explicitly mention that the smooth structure on V' is defined with respect to the chosen collar neighbourhoods ι and $\mathrm{id}_{E\times[1,\infty)}$, such that h is indeed smooth. Note that the metrics on both $E \times [1, \infty)$ and DE are complete, therefore h is also complete. Furthermore, g_E is a PSC-metric, hence the scalar curvature of $g_E + dt^2$ is uniformly positive by compactness of E and Example 2.2 (iv). The same holds for the scalar curvature of g. Consequently, scal_h is uniformly positive as well. Now consider the map $p_0\colon E \times [1, \infty) \to M$ defined as the compostion of the projection $E \times [1, \infty) \to E$ and the bundle projection $p\colon E \to M$. Together with the bundle projection $q\colon DE \to M$ we set

$$\pi := p_0 \cup_E q\colon V' \to M,$$

which we will show to be a rank 2 vector bundle. However, in general we can only expect π to be a continous map as the collar neighbourhood ι contains no information about the bundle structure on DE. To resolve this subtlety, we proceed as follows, being quite precise about the details. Choose trivialisations

$$\phi_\alpha\colon U_\alpha \times S^1 \to p^{-1}(U_\alpha),\ \ \phi_\beta\colon U_\beta \times S^1 \to p^{-1}(U_\beta)$$

of $E \to M$ with smooth transition function $\tau_{\alpha\beta}\colon U_\alpha \cap U_\beta \to S^1$. By definition there are then trivialisations

$$\psi_\alpha\colon U_\alpha \times D^2 \to q^{-1}(U_\alpha),\ \ \psi_\beta\colon U_\beta \times D^2 \to q^{-1}(U_\beta)$$

of $DE \to M$ with the same transition function $\tau_{\alpha\beta}$. Let $B \subseteq \mathbb{R}^2$ be the open ball of radius 1 and consider the diffeomorphism

$$\mu\colon \mathbb{R}^2 - B \to S^1 \times [1, \infty),\ w \mapsto \left(\frac{w}{|w|}, |w|\right),$$

which is equivariant under the canonical S^1-actions. Then we can define a trivialisation

$$\theta_\alpha\colon U_\alpha \times (\mathbb{R}^2 - B) \xrightarrow{\mathrm{id}\times\mu} U_\alpha \times S^1 \times [1, \infty) \xrightarrow{\phi_\alpha\times\mathrm{id}} p^{-1}(U_\alpha) \times [1, \infty) = p_0^{-1}(U_\alpha)$$

of $E \times [1, \infty)$ and similiarly $\theta_\beta\colon U_\beta \times (\mathbb{R}^2 - B) \to p_0^{-1}(U_\beta)$. By construction, the transition function of these trivialisations is again $\tau_{\alpha\beta}$. The maps θ_α and ψ_α now glue together to a *homeomorphism*

$$\xi_\alpha\colon U_\alpha \times \mathbb{R}^2 = (U_\alpha \times (\mathbb{R}^2 - B)) \cup (U_\alpha \times D^2) \xrightarrow{\theta_\alpha\cup\psi_\alpha} p_0^{-1}(U_\beta) \cup q^{-1}(U_\beta) = \pi^{-1}(U_\alpha).$$

Likewise we obtain a homeomorphism $\xi_\beta\colon U_\beta \times \mathbb{R}^2 \to \pi^{-1}(U_\beta)$. From the discussion about the transition functions we conclude

$$(\xi_\beta^{-1} \circ \xi_\alpha)(x, w) = (x, \tau_{\alpha\beta}(x) \cdot w).$$

Observe that this map is smooth. Therefore, the collection of maps constructed in the same manner as ξ_α define a smooth structure on V'. This structure may not agree with the one previously defined with respect to the collar neighbourhoods, so for clarity let us refer to the topological manifold V' equipped with this new smooth structure as V. Then the map $\pi\colon V \to M$ is clearly smooth and carries the structure of a *smooth* rank 2 vector bundle. To now obtain a complete metric on V with uniformly positive scalar curvature we recall the following. The smooth structure on V' (defined via the collar neighbourhoods) is up to diffeomorphism the unique smooth structure on the topological manifold V', such that the topological embeddings

$$E \times [1, \infty) \to V', \;\; DE \to V'$$

are smooth embeddings [hcbord, Theorem 1.4]. Note that the inclusion $E \times [1, \infty) \to V$ is locally given by the inlcusion

$$U_\alpha \times (\mathbb{R}^2 - B) \to U_\alpha \times \mathbb{R}^2$$

with respect to the smooth trivialisations θ_α and ξ_α. Similiarly the inclusion $DE \to V$ is locally given by the inlcusion

$$U_\alpha \times D^2 \to U_\alpha \times \mathbb{R}^2$$

with respect to ψ_α and ξ_α. Hence we conclude from the uniqueness statement above that there exists a diffeomorphism $f \colon V \to V'$. Then the metric f^*h on V is complete and of uniformly positive scalar curvature.

5 Further Questions

5.1 Problems with Extending the Construction to the Spin Case

The base spaces M of the circle bundles $E \to M$ constructed in the previous Chapter are all totally non-spin manifolds. As we have seen in Chapter 2 many constructions in positive scalar curvature geometry require the existence of a spin structure, so it may also be desirable to get examples of such circle bundles over spin manifolds. However, our methods do not straightforwardly extend to the spin case. Let us discuss the difficulties that arise.

Suppose we are given an S^1-principal bundle $\pi\colon E \to M$ over a spin manifold M (in particular M is orientable). First of all we cannot conclude the existence of a PSC-metric on E from Theorem 2.11 anymore, since the total space E is itself a spin manifold. The reason is the following. There exists a splitting

$$TE \cong HE \oplus VE,$$

of the tangent bundle TE of E into a *vertical* bundle VE and a *horizontal* bundle HE. The bundle VE is simply the kernel of the differential $d\pi\colon TE \to TM$ and we can then choose HE as any complementary subbundle (for example by choosing a fiber metric on TE). By construction $d\pi\colon HE \to TM$ is an isomorphism on fibers, such that $HE \cong \pi^*TM$. In our case VE is one-dimensional, such that $w_2(VE) = 0$ and therefore

$$w_2(TE) = w_1(HE) \smile w_1(VE) + w_2(HE) = \pi^*w_1(TM) \smile w_1(VE) + \pi^*w_2(TM) = 0.$$

So instead we are inclined to use the spin analogue of Theorem 2.11 to conclude the existence of a PSC-metric on E, namely:

Theorem 2.10. *Let W^{n+1} be a cobordism from M_0 to M_1, $n \geq 5$, such that W is spin. Suppose that M_1 is connected, the inlcusion $M_1 \to W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W, which is of product form near the boundary.*

Proceeding as before we would like to take the disk bundle DE as a nullbordism of E and show:

(I′) DE is spin,

(II′) The inclusion $S^1 \to E$ is trivial on π_1.

However, these properties contradict themselves. To see this we make the following observations. Let $L \to M$ be the complex line bundle associated to $E \to M$ and consider the inclusion $\iota\colon DE \to L$. Since the differential $d\iota\colon TDE \to TL$ is an isomorphism on fibers we conclude $TDE \cong \iota^*TL$. Thus

$$\iota^*w_2(TL) = w_2(TDE) = 0$$

by property (I′). But ι is a homotopy equivalence so consequently $w_2(TL) = 0$. Let $p\colon L \to M$ be the bundle proejction. As before we have a splitting

$$TL \cong HL \oplus VL$$

into a vertical and horizontal subbundle with $HL \cong p^*TM$. For vector bundles it is a standard fact that $VL = \ker(dp) \cong p^*E$, hence

$$0 = w_2(TL) = \pi^*w_2(HL) + \pi^*w_1(HL) \smile p^*w_1(L) + p^*w_2(L) = p^*w_2(L).$$

Again we conclude $w_2(E) = 0$ since p is a homotopy equivalence. Therefore the first chern class $c_1(L)$ lies in the kernel of the coefficient homomorphism

$$H^2(M;\mathbb{Z}) \to H^2(M;\mathbb{Z}_2)$$

by Proposition 1.17. In other words, $c_1(L)$ can be written as $c_1(L) = 2\alpha$ for some $\alpha \in H^2(M;\mathbb{Z})$. To derive a contradiction from this we claim that the following diagram commutes up to sign

$$\begin{tikzcd} \pi_2(M) \arrow[r, "\partial"] \arrow[d] & \pi_1(S^1) \arrow[d, "\cong"] \\ H_2(M;\mathbb{Z}) \arrow[r, "\langle c_1(L), - \rangle [S^1]"] & H_1(S^1;\mathbb{Z}) \end{tikzcd}$$

where ∂ is the boundary map from the long exact sequence of homotopy groups for the fibration $E \to M$ and the vertical maps are given by the Hurewicz homomorphism. Moreover $[S^1]$ denotes the fundamental class of S^1. Granting this claim for now, it then follows from $c_1(L) = 2\alpha$ that $\partial\colon \pi_2(M) \to \pi_1(S^1)$ cannot be surjective and consequently $\pi_1(S^1) \to \pi_1(E)$ is non-zero by exactness. Hence a bundle satisfying both (I′) and (II′) does not exist. To show commutativity of the above square let $\tau_1 \to \mathbb{C}P^\infty$ be the universal complex line bundle (compare Example 1.14 (vi)) and let $f\colon M \to \mathbb{C}P^\infty$ be a classifiying map for $L \to M$. Then essentially commutativity is due to the fact that the corresponding square for the *universal* principal bundle commutes and all maps are compatible with the classifiying map f. To make this precise we extend the diagram as follows

$$\begin{tikzcd}
\pi_2(\mathbb{C}P^\infty) \arrow[rrr, bend left=5, "\partial", "\cong"'] \arrow[ddd, "\cong"'] & & & \pi_1(S^1) \arrow[ddd, "\cong"] \\
& \pi_2(M) \arrow[r, "\partial"] \arrow[ul, "f_*"'] \arrow[d] & \pi_1(S^1) \arrow[ur, "\mathrm{id}"'] \arrow[d, "\cong"] & \\
& H_2(M;\mathbb{Z}) \arrow[r, "\langle c_1(L),-\rangle[S^1]"] \arrow[dl, "f_*"] & H_1(S^1;\mathbb{Z}) \arrow[dr, "\mathrm{id}"] & \\
H_2(\mathbb{C}P^\infty;\mathbb{Z}) \arrow[rrr, bend right=5, "\langle c_1(\tau_1),-\rangle[S^1]"] & & & H_1(S^1;\mathbb{Z})
\end{tikzcd}$$

Again the top horizontal map is the boundary map from the long exact sequence for the fibration $ES^1 \to \mathbb{C}P^\infty$ and the outer vertical maps are given by the Hurewicz homomorphism. Since $\mathbb{C}P^\infty$ is simply connected $\pi_2(\mathbb{C}P^\infty) \to H_2(\mathbb{C}P^\infty;\mathbb{Z})$ is an isomorphism. Moreover, the boundary map $\partial\colon \pi_2(\mathbb{C}P^\infty) \to \pi_1(S^1)$ is an isomorphism by Example 1.14 (ii). Clearly 1 commutes. Furthermore 2 commutes by naturality of the long exact sequence of homotopy groups and 3 commutes by naturality of the Hurewicz homomorphism. For commutativity of 4 we compute for $a \in H_2(M;\mathbb{Z})$:

$$\langle c_1(\tau_1), f_*a\rangle = \langle f^*c_1(\tau_1), a\rangle = \langle c_1(L), a\rangle,$$

using the fact that f is a classifiying map in the last equation. Finally we have to show commutativity of the outer square. From the universal coefficient theorem we obtain that

$$H^2(\mathbb{C}P^\infty;\mathbb{Z}) \to \mathrm{Hom}(H_2(\mathbb{C}P^\infty;\mathbb{Z}),\mathbb{Z}), \quad \beta \mapsto \langle \beta, -\rangle$$

is an isomorphism. By definiton $c_1(\tau_1) \in H^2(\mathbb{C}P^\infty;\mathbb{Z})$ is a generator, thus $\langle c_1(\tau_1), -\rangle$ is a generator of $\mathrm{Hom}(H_2(\mathbb{C}P^\infty;\mathbb{Z}),\mathbb{Z})$ and consequently evaluates to ± 1 on a generator of $H_2(\mathbb{C}P^\infty;\mathbb{Z})$. From this we see that

$$\langle c_1(\tau_1), -\rangle[S^1]\colon H_2(\mathbb{C}P^\infty;\mathbb{Z}) \to H_2(S^1;\mathbb{Z})$$

is an isomorphism. So all maps in the outer square are isomorphisms and, moreover, all groups are isomorphic to $\mathbb{Z}$. Hence the outer square commutes up to sign and the claim follows.

5.2 A Question about the Homotopy Groups of the Space of PSC-Metrics

A Appendix

A.1 A Short Introduction to Surgery

Surgery Theory provides a way to change the topology of a manifold in a controlled way. The basic idea is the simple observation, that a product $S^k \times S^{n-k-1}$ of spheres can be considered as the boundary of both $S^k \times D^{n-k}$ and of $D^{k+1} \times S^{n-k-1}$. This allows for the following construction. Let M^n be a smooth n-dimensional manifold and let $F\colon S^k \times D^{n-k} \to M$ be an embedding into the interior of M. Then we can remove the interior of the image of F resulting in a manifold

$$\tilde{M} := M - \mathrm{int}(\mathrm{im}(F)).$$

To $\tilde{M}$ we can attach $D^{k+1} \times S^{n-k-1}$ along the boundary component $F(S^k \times S^{n-k-1}) \subseteq \tilde{M}$ to obtain a manifold M'. More precisely we define M' as the pushout

$$\begin{tikzcd} S^k \times S^{n-k-1} \arrow[r, "F|_{S^k \times S^{n-k-1}}"] \arrow[d, "\text{incl.}"] & \tilde{M} \arrow[d] \\ D^{k+1} \times S^{n-k-1} \arrow[r] & M' \end{tikzcd}$$

Up to diffeomorphism there exists a unique smooth structure on M', such that the topological embeddings $\tilde{M} \to M'$ and $D^{k+1} \times S^{n-k-1} \to M'$ are smooth embeddings, see [Milnor1965, Theorem 1.4]. We say that the manifold M' is obtained from M by surgery along F. Intuitively we remove a thickened k-sphere from M and glue in its place a thickened $(n-k-1)$-sphere. Accordingly we refer to the surgery along F as a surgery of dimension k or simply a k-surgery.

A different, yet closely related concept, is the theory of cobordisms. Two closed n-dimensional manifolds M and M' are called *cobordant* if there exists a compact manifold W^{n+1} whose boundary is the disjoint union of M and M'. The manifold W is called a cobordism from M to M'. Since any closed manifold is cobordant to itself via the trivial cobordism $M \times [0,1]$ and cobordisms can be glued together along common boundary components, the relation of cobordism defines an equivalence relation on the class of closed n-dimensional manifolds. One way to construct a new cobordism out of a given one is by means of *handle attachments*. Suppose we are given an embedding $\phi\colon S^k \times D^{n-k} \to \partial W$. Then we can form a new manifold W' by attaching a so called $(k+1)$-handle $D^{k+1} \times D^{n-k}$ to W along ϕ, that is we form the pushout

$$\begin{tikzcd} S^k \times D^{n-k} \arrow[r, "\phi"] \arrow[d, "\text{incl.}"] & W \arrow[d] \\ D^{k+1} \times D^{n-k} \arrow[r] & W' \end{tikzcd}$$

We also refer to $(k+1)$ as the *index* of the handle. Again some care needs to be taken to define a smooth structure on W', see [Milnor, Morse Theory]. It turns out that any cobordism W from M to M' can be obtained from the trivial cobordism $M \times [0,1]$ by attaching finitely many handles [REFERENCE]. Now the connection between surgery theory and cobordisms is the following fundamental result, which can be shown by means of Morse theory.

Theorem A.1 (Milnor1965, Theorem 3.13 and Theorem 3.14 + other source!!)**.** *Let M and M' be closed manifolds of the same dimension. Then M' is obtained from M by a surgery of dimension k if and only if there exists a cobordism W from M to M' containing only a single handle of index $k+1$, that is W is obtained from the trivial cobordism $M \times [0,1]$ by attaching a single $(k+1)$-handle.*

One common usecase of surgery is to modify a given manifold M in order to simplify its homotopy groups. Essentially given some embedding $f\colon S^k \to M$ we would like to modify the topology of M, such that f becomes contractible. We could just glue in a disk D^{k+1} along f, however this would destroy the manifold structure. So instead we "thicken" the embedded sphere by extending f to an embedding $F\colon S^k \times D^{n-k}$, remove this thickened sphere and glue in a whole family of disks $D^{k+1} \times S^{n-k-1}$, i.e. we perform a k-surgery. This process is usually referred to as "killing" the homotopy class $[f]$ by surgery. The fundamental osbtruction to performing such a surgery is the existence of the embedding F, which exists if and only if the normal bundle of f is trivial. The following Proposition now formalizes this idea of killing homotopy classes by surgery.

Figure 2: Test

Proposition A.2. *Let W^n be a smooth manifold with basepoint $w_0 \in W$, $M \subseteq \partial W$ a subset of the boundary with basepoint $x_0 \in M$, $2(k+1) \leq n$ and $k \geq 1$.*

(i) Let $\alpha_1, \ldots, \alpha_l \in \pi_k(W, w_0)$ be classes, which are represented by embeddings with trivial normal bundle. Then there exists a manifold W' obtained from W by l-many k-surgeries away from the point w_0 together with a homomorphism $\phi\colon \pi_1(W, w_0) \to \pi_1(W', w_0)$ and a ϕ-equivariant epimorphism

$$\pi_k(W, w_0) \twoheadrightarrow \pi_k(W', w_0)$$

containing $\alpha_1, \ldots, \alpha_l$ in its kernel. Moreover,

$$\pi_i(W, w_0) \cong \pi_i(W', w_0)$$

for all $i \leq k-1$.

(ii) Let $\alpha_1^{\mathrm{rel}}, \ldots, \alpha_l^{\mathrm{rel}} \in \pi_k(W, M, x_0)$ be classes, which lift to classes $\alpha_1, \ldots, \alpha_l \in \pi_k(W, x_0)$ under the map $\pi_k(W, x_0) \to \pi_k(W, M, x_0)$ induced by the inclusion. Again assume that $\alpha_1, \ldots, \alpha_l$ are represented by embeddings with trivial normal bundle. Then there exists a manifold W' obtained from W by l-many k-surgeries together with a $\pi_1(M, x_0)$-equivariant epimorphism

$$\pi_k(W, M, x_0) \twoheadrightarrow \pi_k(W', M, x_0)$$

containing $\alpha_1^{\mathrm{rel}}, \ldots, \alpha_l^{\mathrm{rel}}$ in its kernel. Moreover,

$$\pi_i(W, M, x_0) \cong \pi_i(W', M, x_0)$$

for all $1 \leq i \leq k-1$.

Remark A.3. If $\pi_k(W, w_0)$ is abelian (for example if $k \geq 2$) equivariance of the epimorphism $\pi_k(W, w_0) \twoheadrightarrow \pi_k(W', w_0)$ implies that it contains the entire $\mathbb{Z}[\pi_1(W, w_0)]$-module generated by $\alpha_1, \ldots, \alpha_l$ in its kernel. An analogous statement holds in the relative case.

In the proof of the Proposition we will make use of the following basic lemma.

Lemma A.4. *Let X be a topological space and let $H\colon S^k \times [0,1] \to X$ be a homotopy from some map $f\colon S^k \to X$ to some map $g\colon S^k \to X$. Fix a basepoint $s_0 \in S^k$ and consider the path $\gamma(t) := H(s_0, t)$ from $x_0 := f(s_0)$ to $x_1 := g(s_0)$. Then the change of basepoint isomorphism*

$$b_\gamma\colon \pi_k(X, x_0) \to \pi_k(X, x_1)$$

induced by γ maps the class $[f]$ to the class $[g]$.

Proof. Represent the classes $[f]$ and $[g]$ by maps $f'\colon D^k \to X$ and $g'\colon D^k \to X$, which are constant on the boundary. Let $H'\colon D^k \times [0,1] \to X$ be the corresponding homotopy from f' to g', such that $H'(S^{k-1}, t) = \gamma(t)$. Then by $b_\gamma([f'])$ is by definition represented by

$$D^k \to X, \quad y \mapsto \begin{cases} f'(2y), & \|y\| \leq \dfrac{1}{2}, \\ \gamma(2\|y\| - 1), & \|y\| \geq \dfrac{1}{2}. \end{cases}$$

This map is homotopic to g' relative S^{k-1} by virtue of the following homotopy.

$$D^k \times [0,1] \to X, \quad y \mapsto \begin{cases} H'\left(\dfrac{2}{1+t} y, t\right), & \|y\| \leq \dfrac{1}{2}(1+t), \\ \gamma(2\|y\| - 1), & \|y\| \geq \dfrac{1}{2}(1+t). \end{cases}$$

□

Proof of Proposition A.2. We first prove the absolute version **??** of the Proposition in three steps.

Step 1. We start by explaining how to kill a single homotopy class by surgery. Suppose that $w_0 \in W$ is an interior point. Let $\beta \in \pi_k(W, w_0)$ be a class represented by an embedding $f\colon (S^k, s_0) \to (W, w_0)$ for a specified basepoint $s_0 \in S^k$, which extends to an embedding $F\colon S^k \times D^{n-k} \to W$ into the interior of W. Let

$$\tilde{W} := W - \operatorname{int}(\operatorname{im}(F))$$

and define W' by the pushout

$$\begin{array}{ccc} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ D^{k+1} \times S^{n-k-1} & \longrightarrow & W' \end{array}$$

as discussed above. Since the inclusion $S^k \times S^{n-k-1} \to D^{k+1} \times S^{n-k-1}$ is a cofibration it follows that the above suqare is a homotopy pushout, hence the map $\tilde{W} \to W'$ has the same connectivity as $S^k \times S^{n-k-1} \to D^{k+1} \times S^{n-k-1}$, which is k-connected. Similiarly, the pushout

$$\begin{array}{ccc} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ S^k \times D^{n-k} & \longrightarrow & W \end{array}$$

implies that $\tilde{W} \to W$ is $n-k-1$-connected. Choose any point $d_0 \in \partial D^{n-k} = S^{n-k-1}$ and set $w_0' := F(s_0, d_0)$. We now construct an epimorphism $\Psi_k\colon \pi_k(W, w_0) \to \pi_k(W', w_0')$ mapping the class β to 0. Choose a path $\lambda\colon [0,1] \to D^{n-k}$ from the center point $0 \in D^{n-k}$ to d_0 and let

$$\eta(t) := F(s_0, \lambda(t)).$$

Then η is a path from w_0 to w_0' and we can consider the change of basepoint isomorphism

$$b_\eta\colon \pi_k(W, w_0) \to \pi_k(W, w_0')$$

induced by η. From the preceeding arguments it follows, that the map $\pi_i(\tilde{W}, w_0') \to \pi_i(W, w_0')$ induced by the inclusion is an isomorphism for $i \leq k$, since $2(k+1) \leq n$. Thus we can define maps

$$\Psi_i\colon \pi_i(W, w_0) \xrightarrow{b_\eta} \pi_i(W, w_0') \xleftarrow{\cong} \pi_i(\tilde{W}, w_0') \to \pi_i(W', w_0').$$

By the above discussion Ψ_i is an isomorphism for $i \leq k-1$ and an epimorphism for $i = k$. We now show that Ψ_k maps the class β to 0. Define

$$\tilde{f}\colon (S^k, s_0) \to (W, w_0'),\ y \mapsto F(y, d_0).$$

Then $\tilde{f}$ is freely homotopic to f via the homotopy $(y,t) \mapsto F(y, \lambda(t))$, so it follows from Lemma A.4 that $b_\eta(\beta)$ is represented by $\tilde{f}$. Note that the image of $\tilde{f}$ lies in $\tilde{W}$ and thus $\Psi_k(\beta)$ is represented by

$$(S^k, s_0) \to (W', w_0'),\ y \mapsto \tilde{f}(y).$$

This map factors as the following composition of inclusions

$$S^k \to D^{k+1} \times \{d_0\} \to W',$$

which is clearly contractible. Therefore $\Psi_k(\beta) = 0$.

Step 2. Next we explain how to kill multiple homotopy classes at once. To this end assume that W is connected. Let $\beta_i \in \pi_k(W, w_i)$ for interior points $w_i \in W$, $1 \leq i \leq l$. Again suppose that the β_i can be represented by embeddings $f_i\colon (S^k, s_0) \to (W, w_i)$, which extend to embeddings $F_i\colon S^k \times D^{n-k} \to W$ into the interior of W. Moreover assume these embeddings to be disjoint, that is $\operatorname{im}(F_i) \cap \operatorname{im}(F_j) = \emptyset$ for $i \neq j$. This allows us to do surgery along all F_i at once, i.e. we can define

$$\tilde{W} := W - \bigcup_{i=1,\dots,l} \operatorname{int}(\operatorname{im}(F_i))$$

and let W' be defined by the pushout

$$\begin{array}{ccc} \coprod_{i=1,\dots,l} S^k \times S^{n-k-1} & \longrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ \coprod_{i=1,\dots,l} D^{k+1} \times S^{n-k-1} & \longrightarrow & W' \end{array}$$

In order to make sense of the statement, that we can kill all homotopy classes β_i at once we would like to consider them as elements in the same group. To achieve this let $d_0 \in S^{n-k-1}$ and $\lambda\colon [0,1] \to D^{n-k}$ be as in Step 1 and set

$$w_i' := F_i(s_0, d_0), \;\; \eta_i(t) := F_i(s_0, \lambda(t)).$$

Moreover choose any point $v_0 \in \tilde{W}$ together with paths $\gamma_i'\colon [0,1] \to \tilde{W}$ from $w_i' \to v_0$. Such paths exist, since W was assumed to be connected and removing (thickened) spheres of codimension at least 2 preserves connectedness. Let

$$\gamma_i := \gamma_i' * \eta_i\colon [0,1] \to W$$

denote the composed path from w_i to v_0 and consider the following commutative diagram for $j \leq k$.

$$\begin{array}{ccccccc} \pi_j(W, v_0) & \xrightarrow{\text{id}} & \pi_j(W, v_0) & \xleftarrow{\cong} & \pi_j(\tilde{W}, v_0) & \longrightarrow & \pi_j(W', v_0) \\ {\scriptstyle b_{\gamma_i}}\uparrow\cong & & {\scriptstyle b_{\gamma_i'}}\uparrow\cong & & {\scriptstyle b_{\gamma_i'}}\uparrow\cong & & {\scriptstyle b_{\gamma_i'}}\uparrow\cong \\ \pi_j(W, w_i) & \xrightarrow[\cong]{b_{\eta_i}} & \pi_j(W, w_i') & \xleftarrow{\cong} & \pi_j(\tilde{W}, w_i') & \longrightarrow & \pi_j(W', w_i') \end{array}$$

Denote the map defined as the top horizontal composition by

$$\Phi_j\colon \pi_j(W, v_0) \to \pi_j(W', v_0).$$

The same arguments as in Step 1 show that Φ_j is an isomorphism for $j \leq k-1$ and an epimorphism for $j = k$. Moreover we have seen in Step 1, that for $j = k$, the class β_i is contained in the kernel of the bottom horizontal composition, hence $b_{\gamma_i}(\beta_i)$ is also contained in the kernel of Φ_k. Define

$$\phi := \Phi_1\colon \pi_1(W, v_0) \to \pi_1(W', v_0)$$

and note that all maps in the definition of Φ_k are compatible with the respective π_1-actions, hence Φ_k is ϕ-equivariant.

Step 3. Finally we show how to kill multiple homotopy classes as formulated in **??**. Let $g_i\colon (S^k, s_0) \to (W, w_0)$ be embeddings with trivial normal bundle representing the class α_i. By a general position argument we can find isotopies $H_i\colon S^k \times [0,1] \to W$ from g_i to embeddings $f_i\colon S^k \to W - \{w_0\}$, such that $f_i(S^k) \cap f_j(S^k) = \emptyset$ for $i \neq j$. Moreover we may assume, that the image of f_i is contained in the interior of W. Since the normal bundle of g_i is trivial, the same holds for f_i. Therefore we can extend the f_i to embeddings

$$F_i\colon S^k \times D^{n-k} \to W - \{w_0\}$$

into the interior of W, such that $\mathrm{im}(F_i) \cap \mathrm{im}(F_j) = \emptyset$ for $i \neq j$. Now we are in the setting as in Step 2 for $w_i = f_i(s_0)$ and $\beta_i = [f_i] \in \pi_k(W, w_i)$. Define paths $\gamma_i(t) := H_i(s_0, 1-t)$ from w_i to w_0. Then $b_{\gamma_i}(\beta_i) = \alpha_i$ by Lemma A.4. We would like to apply Step 2 to $v_0 = w_0$, however the paths γ_i are not of the same form as in Step 2 yet. Let us ignore this issue for a moment and see how Step 3 follows. Again let W' be the manifold obtained by surgery along $F_1, \dots, F_l$ as before. Step 2 now yields a homomorphism $\phi\colon \pi_1(W, w_0) \to \pi_1(W', w_0)$ together with a ϕ-equivariant epimorphism

$$\Phi_k\colon \pi_k(W, w_0) \to \pi_k(W', w_0)$$

containing the $b_{\gamma_i}(\beta_i) = \alpha_i$ in its kernel. Moreover, $\pi_j(W, w_0) \cong \pi_j(W', w_0)$ for $j \leq k-1$.

We now construct a homotopy to bring γ_i into the desired form. Let $t_i := \sup\{t \in [0,1] \mid \gamma_i([0,t]) \subseteq \mathrm{im}(F_i)\}$ be the first time at which γ_i exits $\mathrm{im}(F_i)$. For $t \in [0, t_i]$ we can then write $\gamma_i(t)$ in the form

$$\gamma_i(t) = F_i(\tilde{\sigma}_i(t), \tilde{\lambda}_i(t)).$$

for suitable paths $\tilde{\sigma}_i, \tilde{\lambda}_i$. Now $\tilde{\sigma}_i$ and $\tilde{\lambda}_i$ are homotopic to the reparametrized paths

$$\sigma_i(t) = \begin{cases} \tilde{\sigma}_i(0) = s_0, & t \in \left[0, \dfrac{t_i}{2}\right] \\ \tilde{\sigma}_i(2t - t_i), & t \in \left[\dfrac{t_i}{2}, t_i\right] \end{cases} \quad \text{and} \quad \lambda_i(t) = \begin{cases} \tilde{\lambda}_i(2t), & t \in \left[0, \dfrac{t_i}{2}\right] \\ \tilde{\lambda}_i(t_i), & t \in \left[\dfrac{t_i}{2}, t_i\right] \end{cases}$$

respectively. So after applying this homotopy we may replace $\tilde{\sigma}_i$ and $\tilde{\lambda}_i$ by σ_i and λ_i. For $t \in \left[0, \frac{t_i}{2}\right]$, $\gamma_i(t)$ can then be written as

$$\gamma_i(t) = F_i(s_0, \lambda_i(t)),$$

which is of the same form as η_i in Step 2. It remains to bring $\gamma_i' := \gamma_i|_{\left[\frac{t_i}{2},1\right]}$ into the desired form, meaning the image of γ_i' should be completely contained within $\tilde{W}$. We can achieve this as follows. Applying a general position argument we can assume, that γ_i' does not intersect any of the embedded spheres $f_1(S^k), \dots, f_l(S^k)$. Consider the deformation retraction

$$r\colon W - \bigcup_{j=1,\dots,l} f_j(S^k) \to \tilde{W}$$

induced by the deformation retractions $S^k \times (D^{n-k} - 0) \to S^k \times S^{n-k-1}$. Then $r \circ \gamma_i'$ is a path in $\tilde{W}$, which is homotopic to γ_i' relative to the endpoints. So in conlcusion we may indeed assume, that γ_i is of the form as in Step 2.

Step 4. We can now deduce the relative version **??** of the proposition using **??** applied to the given classes $\alpha_1, \dots, \alpha_l \in \pi_k(W, x_0)$. Using these classes construct $\tilde{W}$ and W' as before. For $i \leq k$ consider the following diagram, where all maps are induced by inlcusions and the rows are exact.

$$\begin{tikzcd}
\pi_i(M,x_0) \arrow[r] & \pi_i(W,x_0) \arrow[r] & \pi_i(W,M,x_0) \arrow[r] & \pi_{i-1}(M,x_0) \arrow[r] & \pi_{i-1}(W,x_0) \\
\pi_i(M,x_0) \arrow[r] \arrow[u, "\mathrm{id}"] \arrow[d, "\mathrm{id}"] & \pi_i(\tilde{W},x_0) \arrow[r] \arrow[u, "\cong"] \arrow[d] & \pi_i(\tilde{W},M,x_0) \arrow[r] \arrow[u] \arrow[d] & \pi_{i-1}(M,x_0) \arrow[r] \arrow[u, "\mathrm{id}"] \arrow[d, "\mathrm{id}"] & \pi_{i-1}(\tilde{W},x_0) \arrow[u, "\cong"] \arrow[d, "\cong"] \\
\pi_i(M,x_0) \arrow[r] & \pi_i(W',x_0) \arrow[r] & \pi_i(W',M,x_0) \arrow[r] & \pi_{i-1}(M,x_0) \arrow[r] & \pi_{i-1}(W',x_0)
\end{tikzcd}$$

It follows from the Five Lemma that

$$\pi_i(\tilde{W}, M, x_0) \to \pi_i(W, M, x_0)$$

is an isomorphism and

$$\pi_i(\tilde{W}, M, x_0) \to \pi_i(W', M, x_0)$$

is an isomorphism for $i \leq k-1$. This shows $\pi_i(W, M, x_0) \cong \pi_i(W', M, x_0)$ for $i \leq k-1$. If $i = k$ the map $\pi_k(\tilde{W}, x_0) \to \pi_k(W', x_0)$ is surjective and a diagram chase shows that

$$\pi_k(\tilde{W}, M, x_0) \to \pi_k(W', M, x_0)$$

is surjective as well. Thus we can define the epimorphism

$$\Phi_{\mathrm{rel}}\colon \pi_k(W, M, x_0) \xleftarrow{\cong} \pi_k(\tilde{W}, M, x_0) \to \pi_k(W', M, x_0).$$

We have already seen that $\alpha_1, \dots, \alpha_l$ are contained in the kernel of the map

$$\pi_k(W, x_0) \xleftarrow{\cong} \pi_k(\tilde{W}, x_0) \to \pi_k(W', x_0).$$

So commutativity of the above diagram implies

$$\Phi_{\mathrm{rel}}(\alpha_i^{\mathrm{rel}}) = 0.$$

Note that all maps in the definition of Φ_{rel} are $\pi_1(M, x_0)$-equivariant, such that the same holds for Φ_{rel}. $\square$

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